Chapter 1. Introduction and Elementary Concepts

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1 Collection of Survey Data

1.1 Primary and secondary data

Definition 1.1

- **Primary data**: data collected by the investigator directly from the source.
- **Secondary data**: data that has already been collected by some agency or individual and is available in published or unpublished records.

Primary data are collected for the purpose at hand, so they fit the target of the study; secondary data are cheaper but were collected for some other purpose and may not match the target exactly. Primary data can be collected by

- direct personal interviews, or interviews by enumerators,
- questionnaires sent by mail,
- telephone interviews.

1.2 Framing a questionnaire

The person framing the questionnaire should know the field of inquiry in detail. Points to keep in mind:

- The surveyor introduces themselves and states the objective of the survey.
- Questions are clear, unambiguous and to the point; leading questions are avoided.
- Questions affecting the prestige or sentiments of respondents, and questions requiring calculations, are avoided.
- Precise instructions for filling in the questionnaire, including units of measurement, are given.
- The questionnaire is concise. Every question costs response quality, so the number of questions should be kept close to the number of quantities (parameters) one actually wants to estimate.
- The layout is visually appealing.

1.3 Technical terms

Definition 1.2

- An **element** is a unit for which information is sought.
- The **population** (universe) is the aggregate of elements about which inference is to be made.
- **Sampling units** are non-overlapping collections of population elements.
- The **sampling frame** is a list of all sampling units in the population.
- A **sample** is a subset of the population selected from the frame to draw inferences about population characteristics.

Collecting information on every unit of the population is a **complete enumeration** or **census**. Often a census is impossible (the target population is not fully accessible in practice), and a sample survey has advantages over it: greater speed, more detailed information, reduced cost and, perhaps surprisingly, greater accuracy, since a small survey can be run with trained staff and careful follow-up, so non-sampling errors are smaller.

1.4 Probability and non-probability sampling

The method used to select the sample is the **sampling procedure**.

- **Probability sampling**: units are selected by a probability mechanism. Only these procedures are considered in this course.
- **Non-probability sampling**: units are selected without a probability mechanism (convenience, quota, judgement); such procedures are subject to human bias.

1.5 With and without replacement

Definition 1.3

- **With replacement (WR)**: units are drawn one by one, and the unit selected at each draw is returned to the population before the next draw. If order matters, there are N^n possible samples.
- **Without replacement (WOR)**: the selected unit is not returned before the next

draw. There are $\binom{N}{n}$ possible (unordered) samples.

Example 1.4: Birth weights

Four children A, B, C, D have birth weights $5.5, 8.0, 6.5, 7.0$ pounds. The WR samples of size 2 (ordered) are the $4^2 = 16$ pairs

$$(A,A),(A,B),(A,C),(A,D),(B,A),\ldots,(D,D),$$

whereas WOR gives the $\binom{4}{2} = 6$ samples $\{A,B\},\{A,C\},\{A,D\},\{B,C\},\{B,D\},\{C,D\}$.

In R: `sample(n)` is a random permutation of $1,\ldots,n$; `sample(x)` permutes x; `sample(x, n)` draws n items WOR; `sample(x, n, replace=T)` draws WR; `sample(x, replace=T)` is a bootstrap sample.

1.6 Planning and execution of a sample survey

Objectives → population to be studied → sampling unit → sampling frame → sample selection (sample size and procedure) → method of collecting information → pilot survey → handling of nonresponse → organization of fieldwork → analysis of data and report.

1.7 Dense and sparse models

Remark 1.5: Modelling view of sampling designs

Under SRS the data are modelled as $y_i = \mu + \varepsilon_i$ $(i = 1,\ldots,n)$ and $\bar{y} = \frac{1}{n}\sum y_i$ uses every observation for the single parameter μ: a *dense* model. If instead

$$y_i = \mu + \varepsilon_i \ (i = 1,\ldots,n_1), \qquad y_j = \varepsilon_j \ (j = 1,\ldots,n-n_1),$$

only part of the data carries information on μ (a *sparse* model), and estimation must treat the two groups differently. Stratified sampling (Ch. 3) uses $y_i = \mu_1 + \varepsilon_i$ and $y_j = \mu_2 + \varepsilon_j$: every observation is informative, but about its own stratum mean.

2 Elementary Concepts

2.1 Expectation, variance, covariance

If x takes values $x_1,\ldots,x_k$ with probabilities $p_1,\ldots,p_k$,

$$\mathrm{E}(x) = \sum_{i=1}^{k} p_i x_i, \qquad \mathrm{Var}(x) = \mathrm{E}[x - \mathrm{E}(x)]^2 = \mathrm{E}(x^2) - [\mathrm{E}(x)]^2,$$

$$\mathrm{Cov}(x,y) = \mathrm{E}[(x - \mathrm{E}x)(y - \mathrm{E}y)] = \mathrm{E}(xy) - \mathrm{E}(x)\mathrm{E}(y),$$

$$\mathrm{Var}\Big(\sum_{i=1}^{k} a_i x_i\Big) = \sum_{i=1}^{k} a_i^2 \mathrm{Var}(x_i) + 2\sum_{i<j} a_i a_j \mathrm{Cov}(x_i, x_j).$$

2.2 Parameters and statistics

Definition 1.6

A **parameter** is a real-valued function of the variable values of all population units. A **statistic** is a real-valued function of the sample values; a statistic used to estimate a parameter is an **estimator**.

For population values $Y_1, \dots, Y_N$ and sample values $y_1, \dots, y_n$:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} Y_i, \quad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \mu)^2, \quad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i - \mu)^2 = \frac{N}{N-1}\sigma^2,$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2.$$

The finite-population variance S^2 (divisor $N-1$) is the natural target of s^2 under SRS without replacement (Ch. 2), and $(N-1)S^2 = N\sigma^2$.

2.3 Bias, variance and MSE

Definition 1.7

- $\hat{\theta}$ is **unbiased** for θ if $\mathrm{E}(\hat{\theta}) = \theta$. Otherwise its **bias** is $B(\hat{\theta}) = \mathrm{E}(\hat{\theta}) - \theta$ and its **relative bias** is $RB(\hat{\theta}) = B(\hat{\theta})/\theta$.
- Sampling variance $\mathrm{Var}(\hat{\theta}) = \mathrm{E}[\hat{\theta} - \mathrm{E}\hat{\theta}]^2$; standard error $SE(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$.
- Mean squared error $\mathrm{MSE}(\hat{\theta}) = \mathrm{E}(\hat{\theta} - \theta)^2$.
- The **relative efficiency** of $\hat{\theta}_2$ with respect to $\hat{\theta}_1$ is $\mathrm{RE}(\hat{\theta}_1, \hat{\theta}_2) = \mathrm{MSE}(\hat{\theta}_1)/\mathrm{MSE}(\hat{\theta}_2)$; if it exceeds 1, $\hat{\theta}_2$ is more efficient.

Theorem 1.8: Bias–variance decomposition

$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + [B(\hat{\theta})]^2$.

Proof.

$$\begin{aligned}\mathrm{E}(\hat{\theta} - \theta)^2 &= \mathrm{E}\big[(\hat{\theta} - \mathrm{E}\hat{\theta}) + (\mathrm{E}\hat{\theta} - \theta)\big]^2 \\ &= \mathrm{E}(\hat{\theta} - \mathrm{E}\hat{\theta})^2 + (\mathrm{E}\hat{\theta} - \theta)^2 + 2(\mathrm{E}\hat{\theta} - \theta)\,\mathrm{E}(\hat{\theta} - \mathrm{E}\hat{\theta}),\end{aligned}$$

and the last expectation is zero. ∎

2.4 Confidence intervals and the bound on the error

For a desired probability $1-\alpha$ we look for a bound B with

$$P(|\hat{\theta} - \theta| \le B) = 1 - \alpha \iff P(\hat{\theta} - B \le \theta \le \hat{\theta} + B) = 1 - \alpha,$$

so $(\hat{\theta} - B, \hat{\theta} + B)$ is a confidence interval with confidence coefficient $1-\alpha$.

- **Normal case**: if $\hat{\theta}$ is (approximately) normal and unbiased, $B = z_{\alpha/2}\sigma_{\hat{\theta}}$. For $1-\alpha = 0.95$, $z_{0.025} = 1.96 \approx 2$, so $B = 2\sigma_{\hat{\theta}}$, the "2-SD bound on the error of estimation" used throughout the course.
- **Any distribution**: by Chebyshev's inequality $P(|\hat{\theta} - \theta| \le 2\sigma_{\hat{\theta}}) \ge 1 - \frac{1}{4} = 0.75$.

2.5 Sample size determination

Only the investigator can say how much precision is needed. It is expressed as $P(|\hat{\theta}-\theta| \leq B) = 1-\alpha$, where B is the **tolerable error** (margin of error). For surveys of people measuring a proportion, $B = 0.03$ and $\alpha = 0.05$ are typical.

Example 1.9: Opinion polls

With the normal approximation and $p = 1/2$ (the worst case),

$$B = 1.96\sqrt{\frac{p(1-p)}{n}} \leq 1.96\sqrt{\frac{1}{4n}}.$$

Setting $B = 0.03$ gives $n = (1.96)^2/(4{\cdot}0.03^2) = 1067.1$, so about 1,068 respondents; this is why national polls typically report a margin of error of $\pm 3\%$ with about a thousand respondents.

Chapter 2. Simple Random Sampling

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1 Introduction

Definition 2.1

Simple random sampling (SRS) is the procedure in which every possible sample of size n has the same chance of being selected; in particular every population unit has the same chance of being in the sample. Without replacement each of the $\binom{N}{n}$ samples has probability $1/\binom{N}{n}$.

A simple random sample can be drawn by the lottery method or with random number tables / computer programs (`sample(N, n)` in R).

Setting. The population values are $u_1, \dots, u_N$ with

$$\mu = \frac{1}{N}\sum_{i=1}^{N} u_i, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(u_i-\mu)^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(u_i-\mu)^2,$$

and $y_1, \dots, y_n$ is an SRS drawn without replacement (WOR). The **sampling fraction** is $f = n/N$ and $1-f$ is the **finite population correction** (fpc).

2 Estimation of the Population Mean

Theorem 2.2: Sample mean under SRS (WOR)

Let $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2$. Then

(i) $\mathrm{E}(\bar{y}) = \mu$;

(ii) $\mathrm{Var}(\bar{y}) = \dfrac{N-n}{N-1}\dfrac{\sigma^2}{n} = \dfrac{N-n}{N}\dfrac{S^2}{n} = \dfrac{S^2}{n}(1-f)$;

(iii) $\mathrm{E}(s^2) = S^2$, hence $\widehat{\mathrm{Var}}(\bar{y}) = \dfrac{s^2}{n}(1-f)$ is unbiased for $\mathrm{Var}(\bar{y})$.

Derivation: proof of the theorem

Marginal distribution of one draw. By symmetry each y_i equals u_k with probability $1/N$, so

$$\mathrm{E}(y_i) = \sum_{k=1}^{N} u_k \frac{1}{N} = \mu, \qquad \mathrm{Var}(y_i) = \sum_{k=1}^{N} (u_k - \mu)^2 \frac{1}{N} = \sigma^2 \quad \text{for all } i.$$

(i) follows: $\mathrm{E}(\bar{y}) = \frac{1}{n}\sum \mathrm{E}(y_i) = \mu$.

Covariance of two draws. For $i \neq j$ the pair (y_i, y_j) is (u_k, u_l), $k \neq l$, with probability $\frac{1}{N(N-1)}$. Hence

$$\begin{aligned}
\mathrm{Cov}(y_i, y_j) &= \mathrm{E}(y_i y_j) - \mu^2 = \frac{1}{N(N-1)} \sum_{k \neq l} u_k u_l - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\
&= \frac{1}{N(N-1)}\Big[\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\
&= \frac{1}{N(N-1)}\Big[\frac{1}{N}\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] = -\frac{1}{N(N-1)}\Big(\sum_k u_k^2 - N\mu^2\Big) \\
&= -\frac{1}{N(N-1)} \sum_k (u_k - \mu)^2 = -\frac{\sigma^2}{N-1}.
\end{aligned}$$

Variance of the mean.

$$\begin{aligned}
\mathrm{Var}(\bar{y}) &= \frac{1}{n^2}\Big[\sum_i \mathrm{Var}(y_i) + 2\sum_{i<j} \mathrm{Cov}(y_i, y_j)\Big] = \frac{1}{n^2}\Big[n\sigma^2 - n(n-1)\frac{\sigma^2}{N-1}\Big] \\
&= \frac{\sigma^2}{n}\Big(1 - \frac{n-1}{N-1}\Big) = \frac{\sigma^2}{n}\frac{N-n}{N-1} = \frac{S^2}{n}\frac{N-n}{N},
\end{aligned}$$

using $N\sigma^2 = (N-1)S^2$.

Unbiasedness of s^2. Using $\sum (y_i - \bar{y})^2 = \sum (y_i - \mu)^2 - n(\bar{y} - \mu)^2$,

$$\mathrm{E}(s^2) = \frac{1}{n-1}\big[n\sigma^2 - n\mathrm{Var}(\bar{y})\big] = \frac{\sigma^2}{n-1}\Big(n - \frac{N-n}{N-1}\Big) = \frac{\sigma^2}{n-1} \cdot \frac{N(n-1)}{N-1} = \frac{N}{N-1}\sigma^2 = S^2.$$

Therefore $\mathrm{E}[\widehat{\mathrm{Var}}(\bar{y})] = \frac{N-n}{Nn}\mathrm{E}(s^2) = \mathrm{Var}(\bar{y})$.

Remark 2.3: SRS with replacement

If the draws are made with replacement, the y_i are iid with mean μ and variance σ^2. Then $\mathrm{Cov}(y_i, y_j) = 0$ (e.g. for 0/1 values $\mathrm{E}(y_i y_j) = P(y_i = 1)P(y_j = 1) = \frac{N_1}{N}\frac{N_1}{N}$), so

$$\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}, \qquad \mathrm{E}(s^2) = \frac{1}{n-1}\Big(n\sigma^2 - n\frac{\sigma^2}{n}\Big) = \sigma^2, \qquad \widehat{\mathrm{Var}}(\bar{y}) = \frac{s^2}{n}.$$

Without replacement the variance is smaller by the factor $\frac{N-n}{N-1}$; the fpc matters only when f is not small.

Confidence interval and bound. A $100(1-\alpha)\%$ confidence interval for μ is

$$\bar{y} \pm z_{\alpha/2}\sqrt{1-f}\,\frac{s}{\sqrt{n}},$$

and with $\alpha = 0.05$ we use the 2-SD bound $B = 2\sqrt{\widehat{\mathrm{Var}}(\bar{y})} = 2\sqrt{\frac{s^2}{n}(1-f)}$.

Example 2.4: Tractors per village

Estimate the average number of tractors per village; $N = 69$, $n = 10$, $\bar{y} = 16.7$, $s^2 = 81.344$.

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{s^2}{n}\cdot\frac{N-n}{N} = \frac{81.344}{10}\cdot\frac{59}{69} = 6.956, \qquad 2\sqrt{6.956} = 5.27,$$

so the interval is $16.7 \pm 5.27 = [11.43,\ 21.97]$.

3 Estimation of the Population Total

The population total is $\tau = N\mu$, and with N known the natural estimator is $\hat{\tau} = N\bar{y}$.

Theorem 2.5

$\mathrm{E}(\hat{\tau}) = \tau$ and

$$\mathrm{Var}(\hat{\tau}) = N^2\frac{N-n}{N-1}\frac{\sigma^2}{n} = N^2\frac{N-n}{N}\frac{S^2}{n}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2\frac{N-n}{N}\frac{s^2}{n} = N^2\frac{s^2}{n}(1-f).$$

A $100(1-\alpha)\%$ interval is $\hat{\tau} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$, typically $\hat{\tau} \pm 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$.

Example 2.6: Total number of tractors

$\hat{\tau} = 69 \times 16.7 = 1152.3$, $\widehat{\mathrm{Var}}(\hat{\tau}) = 69^2 \times 6.956 = 33{,}117$, and $2\sqrt{33117} = 364.0$. The interval is $1152.3 \pm 364.0 = [788,\ 1516]$.

4 Sample Size for Estimating μ and τ

To estimate μ with a bound B, set $2\sqrt{\mathrm{Var}(\bar{y})} = B$:

$$B = 2\sqrt{\frac{N-n}{N-1}\frac{\sigma^2}{n}} \implies n = \frac{N\sigma^2}{(N-1)D+\sigma^2} = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4}.$$

Derivation

Write $D = B^2/4$. Then $D\,n(N-1) = (N-n)\sigma^2$, i.e. $n[(N-1)D+\sigma^2] = N\sigma^2$. The second form follows from $\sigma^2 = \frac{N-1}{N}S^2$.

For the total, $B = 2\sqrt{N^2\frac{N-n}{N}\frac{S^2}{n}}$ gives

$$n = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4N^2}.$$

In practice S^2 (or σ^2) is unknown and replaced by a prior estimate or a pilot-sample s^2.

Example 2.7: Tractors, continued

With $s^2 = 81.344$ and $B = 3$ ($D = 2.25$),

$$n = \frac{69 \times 81.344}{69 \times 2.25 + 81.344} = \frac{5612.7}{236.6} = 23.72 \;\Rightarrow\; n = 24,$$

so 14 more villages are needed beyond the 10 already sampled.

5 Estimation of a Population Proportion

Let $y_i = 1$ if the ith unit has the characteristic and 0 otherwise. Then the population proportion is the population mean of the 0/1 values, $p = \mu$, and

$$\hat{p} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \hat{q} = 1 - \hat{p}.$$

For 0/1 values $\sigma^2 = pq$ and $S^2 = \frac{N}{N-1}pq$.

Theorem 2.8: Proportion under SRS

$$\mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{pq}{n}\frac{N-n}{N-1}, \qquad \widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N} = \frac{\hat{p}\hat{q}}{n-1}(1-f),$$

and $\mathrm{E}[\widehat{\mathrm{Var}}(\hat{p})] = \mathrm{Var}(\hat{p})$.

Derivation: why the divisor is $n-1$

$$\mathrm{E}(\hat{p}\hat{q}) = \mathrm{E}(\hat{p}) - \mathrm{E}(\hat{p}^2) = p - \left[\mathrm{Var}(\hat{p}) + p^2\right] = pq - \frac{pq}{n}\frac{N-n}{N-1}$$
$$= pq\,\frac{n(N-1) - (N-n)}{n(N-1)} = \frac{N(n-1)}{(N-1)n}\,pq.$$

Hence

$$\mathrm{E}\Big[\frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N}\Big] = \frac{N-n}{N(n-1)} \cdot \frac{N(n-1)}{(N-1)n}pq = \frac{pq}{n}\frac{N-n}{N-1} = \mathrm{Var}(\hat{p}).$$

By contrast the plug-in $\frac{\hat{p}\hat{q}}{n}(1-f)$ is biased downward: $\mathrm{E}\big[\frac{\hat{p}\hat{q}}{n}(1-f)\big] \neq \mathrm{Var}(\hat{p})$. This is the same phenomenon as s^2 versus the divisor-n variance, since $s^2 = \frac{n}{n-1}\hat{p}\hat{q}$ for 0/1 data.

Example 2.9: Student vote

An SRS of $n = 120$ from $N = 2400$ students; 72 support candidate A. Then $\hat{p} = 0.6$ and

$$\widehat{\mathrm{Var}}(\hat{p}) = \Big(1 - \frac{120}{2400}\Big)\frac{0.6 \times 0.4}{119} = 0.001916, \qquad 2\sqrt{0.001916} = 0.0875,$$

so the interval is $0.6 \pm 0.0875 = (0.512,\ 0.688)$.

6 Sample Size for Estimating p

Replace σ^2 by pq in the formula for μ:

$$n = \frac{Npq}{(N-1)D + pq}, \qquad D = \frac{B^2}{4} \quad (\text{or } D = B^2/1.96^2 \text{ with the exact normal quantile}).$$

With no prior information take $p = 1/2$, which maximizes pq and gives a conservative n.

Example 2.10

For $N = 2400$ and $B = 0.05$: $D = 0.05^2/4 = 0.000625$ and

$$n = \frac{2400 \times 0.25}{2399 \times 0.000625 + 0.25} = \frac{600}{1.7494} = 342.98 \;\Rightarrow\; n = 343.$$

7 Towards Stratified Sampling

Consider estimating the total first-hospitalization cost of kidney-stone patients. Two regions with very different incidence rates are surveyed separately:

	Carolinas	Rockies
n	363	258
$\bar{y}$	1350	1150

What is the total cost? Pooling all 621 observations into one mean ignores the fact that the regions were sampled separately and have different population sizes. The answer is to weight each regional mean by its population size, which is the stratified estimator of Chapter 3.

Chapter 3. Stratified Random Sampling

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1 Introduction

Definition 3.1

A **stratified random sample** is obtained by separating the population into non-overlapping groups called **strata** and then selecting a simple random sample from each stratum, independently across strata.

Because the strata are sampled independently, each stratum can be treated as a separate population. Stratification pays off when the variation *between* strata is large (the strata are internally homogeneous). Model-wise, SRS uses $y_i = \mu + \varepsilon_i$, while stratified sampling uses $y_{ij} = \mu_j + \varepsilon_{ij}$: each stratum has its own mean. Once the selection procedure is fixed, the design questions are

- how many strata to construct,
- how to allocate the total sample size to the strata,
- how to choose the strata.

2 Notation

Strata are indexed by $j = 1, \ldots, L$ and units within a stratum by i; y_{ij} is the observation on unit i of stratum j. Sampling within each stratum is SRS without replacement.

N_j, n_j	population and sample size of stratum j; $N = \sum N_j$, $n = \sum n_j$
$w_j = N_j/N$	stratum weight
$f_j = n_j/N_j$	sampling fraction in stratum j
$\mu_j = \frac{1}{N_j}\sum_{i=1}^{N_j} Y_{ij}$	true stratum mean
$\bar{y}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} y_{ij}$	sample stratum mean
$\sigma_j^2 = \frac{1}{N_j}\sum_i (Y_{ij}-\mu_j)^2$	true stratum variance
$S_j^2 = \frac{N_j}{N_j-1}\sigma_j^2$	stratum variance with divisor $N_j - 1$
$s_j^2 = \frac{1}{n_j-1}\sum_{i=1}^{n_j}(y_{ij}-\bar{y}_j)^2$	sample stratum variance

3 Estimation of the Population Mean

Theorem 3.2: Stratified estimator of μ

$$\bar{y}_{\text{st}} = \frac{1}{N}\sum_{j=1}^{L} N_j \bar{y}_j = \sum_{j=1}^{L} w_j \bar{y}_j$$

is unbiased for μ, with

$$\operatorname{Var}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{S_j^2}{n_j}, \qquad \widehat{\operatorname{Var}}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{s_j^2}{n_j}, \qquad \mathrm{E}[\widehat{\operatorname{Var}}(\bar{y}_{\text{st}})] = \operatorname{Var}(\bar{y}_{\text{st}}).$$

A $100(1-\alpha)\%$ interval is $\bar{y}_{\text{st}} \pm z_{\alpha/2}\sqrt{\widehat{\operatorname{Var}}(\bar{y}_{\text{st}})}$.

Proof. Each $\bar{y}_j$ is an SRS mean, so by Ch. 2, $\mathrm{E}(\bar{y}_j) = \mu_j$, $\operatorname{Var}(\bar{y}_j) = (1-f_j)S_j^2/n_j$ and $\mathrm{E}(s_j^2) = S_j^2$. Thus

$$\mathrm{E}(\bar{y}_{\text{st}}) = \sum_j w_j \mu_j = \frac{1}{N}(N_1\mu_1 + \cdots + N_L\mu_L) = \frac{1}{N}(\tau_1 + \cdots + \tau_L) = \frac{\tau}{N} = \mu.$$

Since the strata are sampled independently the covariances vanish and $\operatorname{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \operatorname{Var}(\bar{y}_j)$. Unbiasedness of $\widehat{\operatorname{Var}}$ follows from $\mathrm{E}(s_j^2) = S_j^2$. ∎

4 Estimation of the Population Total

Theorem 3.3

$\hat{\tau}_{\text{st}} = N\bar{y}_{\text{st}} = \sum_{j=1}^{L} N_j \bar{y}_j$ is unbiased for τ, and

$$\operatorname{Var}(\hat{\tau}_{\text{st}}) = N^2 \operatorname{Var}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} N_j^2 (1-f_j)\frac{S_j^2}{n_j}, \qquad \widehat{\operatorname{Var}}(\hat{\tau}_{\text{st}}) = \sum_{j=1}^{L} N_j^2 (1-f_j)\frac{s_j^2}{n_j}.$$

Example 3.4: TV-viewing time

	N_j	w_j	n_j	$\bar{y}_j$	s_j
Town A	155	0.5	20	33.90	5.95
Town B	62	0.2	8	25.12	15.25
Rural	93	0.3	12	19.00	9.36
Total	310		40		

Mean.

$$\bar{y}_{\text{st}} = \frac{155(33.90) + 62(25.12) + 93(19.00)}{310} = \frac{8578.94}{310} = 27.67.$$

$$\begin{aligned}\widehat{\text{Var}}(\bar{y}_{\text{st}}) &= (0.5)^2 \frac{5.95^2}{20}\left(1 - \frac{20}{155}\right) + (0.2)^2 \frac{15.25^2}{8}\left(1 - \frac{8}{62}\right) + (0.3)^2 \frac{9.36^2}{12}\left(1 - \frac{12}{93}\right) \\ &= 0.385 + 1.013 + 0.572 = 1.97,\end{aligned}$$

so the estimate with a 2-SD bound is $27.67 \pm 2\sqrt{1.97} = 27.67 \pm 2.81$.
Total. $\hat{\tau}_{\text{st}} = 310(27.67) = 8579$, $\widehat{\text{Var}}(\hat{\tau}_{\text{st}}) = 310^2(1.97) \approx 189{,}400$, and the bound is $2\sqrt{189{,}400} \approx 870$.

5 Sample Size for Estimating μ and τ

Setting $2\sqrt{\text{Var}(\bar{y}_{\text{st}})} = B$ alone cannot be solved for n; we must also fix how n is split among the strata. Write the allocation as $n_j = n a_j$ with $\sum a_j = 1$.

Theorem 3.5: Sample size for a given allocation

$$n = \frac{\sum_{j=1}^{L} N_j^2 S_j^2 / a_j}{N^2 D + \sum_{j=1}^{L} N_j S_j^2}, \qquad D = \begin{cases} B^2/4 & \text{for } \mu, \\ B^2/(4N^2) & \text{for } \tau, \end{cases}$$

with S_j^2 replaced by prior estimates s_j^2 in practice.

Derivation

For μ let $D = (B/2)^2 = \text{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}(1 - \frac{n_j}{N_j})$. Multiply by N^2 and substitute $n_j = n a_j$:

$$N^2 D = \sum_j N_j^2 \frac{S_j^2}{n a_j} - \sum_j N_j S_j^2 \implies n\Big(N^2 D + \sum_j N_j S_j^2\Big) = \sum_j \frac{N_j^2 S_j^2}{a_j}.$$

For τ, $\text{Var}(\hat{\tau}_{\text{st}}) = N^2 \text{Var}(\bar{y}_{\text{st}}) = B^2/4$ gives the same formula with $D = B^2/(4N^2)$. (Using 1.96 instead of 2 replaces $D = (B/2)^2$ by $(B/1.96)^2$.)

Example 3.6: Equal allocation

For the TV population ($N_j = 155, 62, 93$) with prior variances $s_1^2 = 25$, $s_2^2 = 225$, $s_3^2 = 100$, equal allocation $a_j = 1/3$ and $B = 2$ for μ ($D = 1$):

$$n = \frac{3(155^2 \cdot 25 + 62^2 \cdot 225 + 93^2 \cdot 100)}{310^2(1) + (155 \cdot 25 + 62 \cdot 225 + 93 \cdot 100)} = \frac{6{,}991{,}275}{123{,}225} = 56.7,$$

so $n = 57$ and $n_1 = n_2 = n_3 = 19$.

6 Allocation of the Sample

Let the survey cost be

$$TC = c_0 + \sum_{j=1}^{L} c_j n_j,$$

where c_0 is the overhead cost and c_j the cost per observation in stratum j.

6.1 Equal allocation

$n_j = n/L$. Then $TC = c_0 + \frac{n}{L}\sum c_j$, so $n = \dfrac{(TC - c_0)L}{\sum_j c_j}$.

6.2 Proportional allocation

$n_j = nw_j = nN_j/N$, i.e. $a_j = w_j$ and all $f_j = f$ are equal. Then $TC = c_0 + n\sum c_j w_j$, so $n = \dfrac{TC - c_0}{\sum_j c_j w_j}$; if $c_j \equiv c$, $n = (TC - c_0)/c$. The estimator becomes self-weighting: $\bar{y}_{\text{st}}$ equals the plain mean of all n observations.

6.3 Optimum allocation

Theorem 3.7: Optimum allocation

For a fixed total cost (equivalently, for a fixed variance) $\mathrm{Var}(\bar{y}_{\text{st}})$ is minimized when

$$n_j \propto \frac{w_j S_j}{\sqrt{c_j}}, \qquad \text{i.e.} \qquad n_j = n\,\frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}},$$

and the total sample size under a budget TC is

$$n = \frac{(TC - c_0)\sum_k w_k S_k/\sqrt{c_k}}{\sum_k w_k S_k \sqrt{c_k}}.$$

Derivation: Cauchy–Schwarz argument

Split the variance into the part depending on the allocation and a constant:

$$\mathrm{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\Big(1 - \frac{n_j}{N_j}\Big) = \underbrace{\sum_j \frac{w_j^2 S_j^2}{n_j}}_{V'} - \underbrace{\sum_j \frac{w_j S_j^2}{N}}_{\text{const}}.$$

Let $C' = TC - c_0 = \sum_j c_j n_j$. Minimizing V' for fixed C', or C' for fixed V', is the same as minimizing the product

$$V'C' = \Big(\sum_j \frac{w_j^2 S_j^2}{n_j}\Big)\Big(\sum_j c_j n_j\Big) = \Big(\sum_j a_j^2\Big)\Big(\sum_j b_j^2\Big), \quad a_j = \frac{w_j S_j}{\sqrt{n_j}},\ b_j = \sqrt{c_j n_j}.$$

By the Cauchy–Schwarz inequality

$$V'C' \geq \Big(\sum_j a_j b_j\Big)^2 = \Big(\sum_j w_j S_j \sqrt{c_j}\Big)^2,$$

whose right-hand side does not depend on the n_j. Equality holds iff b_j/a_j is constant:

$$\frac{b_j}{a_j} = \frac{n_j\sqrt{c_j}}{w_j S_j} = \text{const} \iff n_j \propto \frac{w_j S_j}{\sqrt{c_j}}.$$

Substituting $n_j = n\,\frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}}$ into $TC = c_0 + \sum c_j n_j$ gives $TC = c_0 + n\alpha$ with $\alpha = \frac{\sum_k w_k S_k \sqrt{c_k}}{\sum_k w_k S_k/\sqrt{c_k}}$, so $n = (TC - c_0)/\alpha$.

We therefore take a larger sample in a stratum that is (1) larger, (2) more variable, or (3) cheaper to sample.

6.4 Neyman allocation

Theorem 3.8: Neyman allocation

If the costs are equal, $c_j \equiv c$, the optimum allocation becomes

$$n_j = n\,\frac{w_j S_j}{\sum_k w_k S_k} = n\,\frac{N_j S_j}{\sum_k N_k S_k}.$$

Example 3.9: Neyman allocation for the TV data

With $n = 40$ and $N_j s_j = 922.3,\ 945.5,\ 870.5$ (sum 2738.2), the Neyman allocation is

$$n_1 = 13.5, \quad n_2 = 13.8, \quad n_3 = 12.7 \ \Rightarrow\ (13, 14, 13),$$

compared with the proportional allocation $(20, 8, 12)$ actually used. The small but very variable Town B gets many more observations. Treating s_j as S_j, the variance of $\bar{y}_{\text{st}}$ drops from 1.97 (proportional) to about 1.66 (Neyman).

7 Estimation of a Population Proportion

With $y_{ij} \in \{0, 1\}$ the stratum mean $\bar{y}_j$ is the stratum proportion $\hat{p}_j$, and

$$\hat{p}_{\text{st}} = \frac{1}{N}(N_1\hat{p}_1 + \cdots + N_L\hat{p}_L) = \sum_j w_j \hat{p}_j, \qquad \mathrm{E}(\hat{p}_{\text{st}}) = \sum_j w_j p_j = p.$$

Theorem 3.10

$$\mathrm{Var}(\hat{p}_{\text{st}}) = \sum_j w_j^2 \frac{p_j q_j}{n_j}\Big(\frac{N_j - n_j}{N_j - 1}\Big), \qquad \widehat{\mathrm{Var}}(\hat{p}_{\text{st}}) = \sum_j w_j^2 \frac{\hat{p}_j \hat{q}_j}{n_j - 1}(1 - f_j),$$

and a $100(1-\alpha)\%$ interval for p is $\hat{p}_{\text{st}} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{p}_{\text{st}})}$.

These follow from Ch. 2 applied stratum by stratum, the divisor $n_j - 1$ making each term unbiased.

Chapter 4. Systematic Sampling

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1 Introduction

To select n names from a long list, a simple way is to pick names at equal intervals, e.g. every tenth name. If the starting point is random, the result is a systematic sample.

Definition 4.1

A **1-in-k systematic sample with a random start** is obtained by randomly selecting one element among the first k elements of the frame and every kth element thereafter. Here n denotes the sample size and k the number of possible systematic samples.

Advantages over SRS:

- the sample is easy to draw and the field work is less prone to mistakes;
- it can provide more information per unit cost, because the sample is spread evenly over the frame.

2 Selection Methods

2.1 Method A: random start among the first k

Example 4.2: $N = 12$, $k = 3$

Pick a random unit among y_1, y_2, y_3 and then every third unit. If y_2 is picked, the sample is y_2, y_5, y_8, y_{11}. The population is cut into $n = N/k = 4$ rows and one unit is taken from each row; the $k = 3$ possible samples are the columns

#1	#2	#3
y_1	y_2	y_3
y_4	y_5	y_6
y_7	y_8	y_9
y_{10}	y_{11}	y_{12}

In general the sample is $j,\ j+k,\ j+2k,\ \ldots,\ j+(n-1)k$ with j uniform on $\{1,\ldots,k\}$, and each systematic sample has probability $1/k$.

If $N \neq nk$, the samples have unequal sizes. For $N = 12$ and $k = 5$ we have $2 < N/k = 2.4 < 3$: the five samples are (y_1, y_6, y_{11}), (y_2, y_7, y_{12}), (y_3, y_8), (y_4, y_9), (y_5, y_{10}), of size 2 or 3, yet each is chosen with probability $1/5$.

2.2 Method B: random start among all N

Select one unit, say the jth, at random from the whole population, and divide j by k: if the remainder is r, the start is y_r (with $r = 0$ meaning y_k). Then take every kth unit.

Example 4.3: $N = 11$, $k = 3$

If $j = 8$, then $8 = 2 \cdot 3 + 2$, so $r = 2$, the start is y_2 and the sample is (y_2, y_5, y_8, y_{11}). The possible samples are

$$\#1 = (y_1, y_4, y_7, y_{10}), \quad \#2 = (y_2, y_5, y_8, y_{11}), \quad \#3 = (y_3, y_6, y_9).$$

Sample #2 is obtained whenever $j \in \{2, 5, 8, 11\}$, so its probability is $4/11$; similarly #1 has probability $4/11$ and #3 has $3/11$.

Remark 4.4: A versus B

Both methods produce the same collection of systematic samples, but the selection probabilities differ: $1/k$ under Method A and n_i/N (proportional to the sample size) under Method B. Method B requires N to be known. When $N = nk$ the two coincide.

3 Estimation of the Population Mean

Let y_{ij} be the jth element of the ith systematic sample ($i = 1, \ldots, k$, $j = 1, \ldots, n_i$), $\bar{y}_i = \frac{1}{n_i}\sum_j y_{ij}$, and $\bar{y}_{\text{sy}}$ the mean of the sample actually selected.

3.1 Unbiasedness

Case 1 (Method A, $N = nk$).

$$\mathrm{E}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k}(\bar{y}_1 + \cdots + \bar{y}_k) = \frac{1}{k}\frac{1}{n}(y_1 + \cdots + y_N) = \mu,$$

so $\bar{y}_{\mathrm{sy}}$ is unbiased.

Example 4.5

Population $6, 3, 4, 9, 2, 5, 1, 7, 8$ with $k = 3$: the samples are $(6, 9, 1)$, $(3, 2, 7)$, $(4, 5, 8)$ with means $16/3$, $12/3$, $17/3$, whose average is $5 = \mu$.

Case 2 (Method A, $N \neq nk$). For $y_1, \ldots, y_7$ and $k = 3$,

$$\mathrm{E}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}\left[\frac{y_1 + y_4 + y_7}{3} + \frac{y_2 + y_5}{2} + \frac{y_3 + y_6}{2}\right] \neq \frac{y_1 + \cdots + y_7}{7} = \mu$$

in general: $\bar{y}_{\mathrm{sy}}$ is *biased*.

Case 3 (Method B, $N \neq nk$). With probabilities $3/7, 2/7, 2/7$,

$$\mathrm{E}(\bar{y}_{\mathrm{sy}}) = \frac{3}{7}\cdot\frac{y_1 + y_4 + y_7}{3} + \frac{2}{7}\cdot\frac{y_2 + y_5}{2} + \frac{2}{7}\cdot\frac{y_3 + y_6}{2} = \frac{y_1 + \cdots + y_7}{7} = \mu.$$

In general $\mathrm{E}(\bar{y}_{\mathrm{sy}}) = \sum_i \frac{n_i}{N}\bar{y}_i = \frac{1}{N}\sum_i \sum_j y_{ij} = \mu$: Method B is always unbiased.

3.2 Variance of $\bar{y}_{\mathrm{sy}}$ ($N = nk$)

By definition,

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_i - \mu)^2,$$

where $S^2 = \frac{1}{N-1}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \mu)^2$.

Theorem 4.6: Variance of the systematic mean

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{N-1}{N}S^2 - \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_i)^2.$$

Proof. Decompose the total sum of squares:

$$\sum_i\sum_j(y_{ij} - \mu)^2 = \sum_i\sum_j(y_{ij} - \bar{y}_i)^2 + n\sum_i(\bar{y}_i - \mu)^2 + 2\sum_i(\bar{y}_i - \mu)\sum_j(y_{ij} - \bar{y}_i),$$

and the cross term vanishes because $\sum_j(y_{ij} - \bar{y}_i) = 0$. The left side is $(N-1)S^2$ and $n\sum_i(\bar{y}_i - \mu)^2 = nk\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = N\mathrm{Var}(\bar{y}_{\mathrm{sy}})$. Divide by N. ∎

The first term is the variance of the population as a whole; the second is the pooled variation *within* the k systematic samples. The more heterogeneous the units within a systematic sample, the smaller $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ becomes.

Remark 4.7: Periodicity

For the population $1\,2\,3\,4\,5 \mid 1\,2\,3\,4\,5 \mid 1\,2\,3\,4\,5$ with $k = 5$, every systematic sample is constant, e.g. $(2, 2, 2)$. There is no within-sample variation, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ is as large as it can be.

3.3 Intraclass correlation

Homogeneity within a systematic sample is measured by the **intraclass correlation coefficient**

$$\rho = \frac{\mathrm{E}[(y_{ij}-\mu)(y_{ij'}-\mu)]}{\mathrm{E}[(y_{ij}-\mu)^2]}, \qquad j \neq j',$$

where the expectations are averages over the $kn(n-1)/2$ pairs in the same sample and over the N units, respectively.

Theorem 4.8

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{S^2}{n}\cdot\frac{N-1}{N}[1+(n-1)\rho], \qquad \rho = \frac{2}{(n-1)(N-1)S^2}\sum_{i=1}^{k}\sum_{j<j'}(y_{ij}-\mu)(y_{ij'}-\mu).$$

Derivation

$$\begin{aligned}\mathrm{Var}(\bar{y}_{\mathrm{sy}}) &= \frac{1}{k}\sum_i\Big(\frac{1}{n}\sum_j y_{ij}-\mu\Big)^2 = \frac{1}{n^2k}\sum_i\Big(\sum_j(y_{ij}-\mu)\Big)^2\\ &= \frac{1}{n^2k}\Big[\sum_i\sum_j(y_{ij}-\mu)^2 + 2\sum_i\sum_{j<j'}(y_{ij}-\mu)(y_{ij'}-\mu)\Big].\end{aligned}$$

The denominator of ρ is $\frac{1}{N}\sum_i\sum_j(y_{ij}-\mu)^2 = \frac{N-1}{N}S^2$, and the numerator is $\frac{2}{kn(n-1)}\sum_i\sum_{j<j'}(\cdot)(\cdot)$, so

$$2\sum_i\sum_{j<j'}(y_{ij}-\mu)(y_{ij'}-\mu) = kn(n-1)\frac{N-1}{N}S^2\rho = (n-1)(N-1)S^2\rho,$$

which is the stated formula for ρ. Substituting, with $n^2k = nN$,

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{nN}[(N-1)S^2 + (n-1)(N-1)S^2\rho] = \frac{S^2}{n}\frac{N-1}{N}[1+(n-1)\rho].$$

Compare with SRS, $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{S^2}{n}(1-\frac{n}{N})$:

- ρ large and positive (homogeneous systematic samples) $\Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}})$ large;
- ρ small positive or negative (heterogeneous samples) $\Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}})$ small;
- $\rho \approx 0 \Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$. Indeed $\rho = -\frac{1}{N-1}$ reproduces the SRS variance exactly.

3.4 ANOVA form

Arrange the population as an $n \times k$ table whose columns are the systematic samples, with column means $\bar{y}_1, \ldots, \bar{y}_k$ and grand mean μ. Let

$$SS_T = \sum_i\sum_j(y_{ij}-\mu)^2, \quad SS_W = \sum_i\sum_j(y_{ij}-\bar{y}_i)^2, \quad SS_B = n\sum_i(\bar{y}_i-\mu)^2,$$

so $SS_T = SS_W + SS_B$ and $S^2 = SS_T/(N-1)$. Then

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k}\sum_i(\bar{y}_i-\mu)^2 = \frac{SS_B}{N},$$

and equating with the ρ form, $\frac{SS_T}{nN}[1+(n-1)\rho] = \frac{SS_B}{N}$, gives

$$1+(n-1)\rho = \frac{n\,SS_B}{SS_T} \quad \Longrightarrow \quad \rho = 1 - \frac{n}{n-1}\frac{SS_W}{SS_T} \approx 1 - \frac{SS_W}{SS_T} = \frac{SS_B}{SS_T}.$$

A large between-sample sum of squares means a large ρ and a large variance.

Example 4.9: Two orderings of $1,\ldots,9$

Here $\mu = 5$, $S^2 = 60/8 = 7.5$, $n = k = 3$, and the SRS variance is $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{7.5}{3}\cdot\frac{6}{9} = 1.667$.

- Order 1 2 3 | 4 5 6 | 7 8 9: the samples are $(1,4,7),(2,5,8),(3,6,9)$ with means $4,5,6$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(1+0+1) = \frac{2}{3}$ and $\rho = -21/60$. Heterogeneous samples: systematic beats SRS.
- Order 1 4 7 | 2 5 8 | 3 6 9: the samples are $(1,2,3),(4,5,6),(7,8,9)$ with means $2,5,8$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(9+0+9) = 6$ and $\rho = 51/60$. Homogeneous samples: systematic is much worse than SRS.

Check: $\frac{7.5}{3}\cdot\frac{8}{9}(1+2\rho) = \frac{20}{9}(1+2\rho)$ equals $\frac{2}{3}$ for $\rho = -0.35$ and 6 for $\rho = 0.85$.

3.5 Types of population

We want systematic samples with heterogeneous units (small ρ). Which populations produce them?

1. **Random order.** E.g. estimating the average weight of $N = 3000$ SNU freshmen from an alphabetical list. The units in a systematic sample are effectively in random order, $\rho \approx 0$, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$: the systematic sample may be treated as an SRS.
2. **Ordered population.** E.g. farms ordered by area when estimating corn yield. A systematic sample covers the whole range, is heterogeneous, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) < \mathrm{Var}(\bar{y}_{\mathrm{ran}})$. It avoids samples with too many large or too many small farms, so it tends to be more representative than an SRS.
3. **Periodic variation.** E.g. supermarket sales, high on Friday–Saturday and low on Monday–Tuesday. If k matches the period, the units are homogeneous, ρ is large, and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) > \mathrm{Var}(\bar{y}_{\mathrm{ran}})$.

3.6 Estimating the variance

A single systematic sample contains no information about between-sample variation, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ cannot be estimated unbiasedly from it. Assuming a randomly ordered population ($\rho \approx 0$), we use the SRS formula:

$$\bar{y}_{\mathrm{sy}} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{sy}}) = \frac{s^2}{n}\cdot\frac{N-n}{N}.$$

This is conservative for an ordered population and anti-conservative under periodicity.

4 Population Total and Proportion

Assuming a randomly ordered population:

$$\hat{\tau} = N\bar{y}_{\mathrm{sy}}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{sy}}) = N^2\frac{s^2}{n}\frac{N-n}{N};$$

$$\hat{p}_{\mathrm{sy}} = \bar{y}_{\mathrm{sy}} = \frac{1}{n}\sum y_i \ (y_i \in \{0,1\}), \qquad \widehat{\mathrm{Var}}(\hat{p}_{\mathrm{sy}}) = \frac{\hat{p}_{\mathrm{sy}}\hat{q}_{\mathrm{sy}}}{n-1}\cdot\frac{N-n}{N}.$$

5 Selecting the Sample Size

Under the same assumption the SRS formulas apply: for μ with bound B,

$$n = \frac{N\sigma^2}{(N-1)D + \sigma^2} = \frac{NS^2}{ND + S^2}, \qquad D = \frac{B^2}{4},$$

and for p, $n = \dfrac{Npq}{(N-1)D + pq}$ with $D = B^2/4$.

Chapter 5. Ratio Estimation

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1 Auxiliary Information

So far the data were modelled as $y_i = \mu + \varepsilon_i$ with $\mathrm{E}(\varepsilon_i) = 0$, so $\mathrm{E}(y_i) = \mu$ for every unit. If an auxiliary variable x related to y is available for each sampled unit, a better model is

$$y_i = f(x_i) + \varepsilon_i, \qquad \mathrm{E}(y_i) = f(x_i) \quad (\text{e.g. } f(x) = \beta_0 + \beta_1 x),$$

and knowledge of the population mean μ_x or total τ_x of x can be used to sharpen the estimate of μ_y or τ_y. This chapter uses a line through the origin (ratio estimation); Ch. 7 uses a general line (regression estimation).

Example 5.1: Sugar in a truckload of oranges

The wholesale price of oranges is based on the total sugar content τ_y of the load. The estimate $N\bar{y}$ needs the number of oranges N, which is unknown. But sugar content y is closely related to weight x, and the total weight τ_x of the truckload is easy to obtain. Since

$$\frac{\mu_y}{\mu_x} = \frac{N\mu_y}{N\mu_x} = \frac{\tau_y}{\tau_x} \quad \Longrightarrow \quad \tau_y = \frac{\mu_y}{\mu_x}\tau_x,$$

the **ratio estimator** of the total is

$$\hat{\tau}_y = \frac{\bar{y}}{\bar{x}}\tau_x = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}\tau_x,$$

which does not require N.

Other ratios of interest: total monthly food budget to total monthly income per family; number of children to number of residents; average amount of a new drug needed to evoke a response relative to that of a standard drug.

2 Ratio Estimation under SRS

Let (x_i, y_i), $i = 1, \dots, n$, be an SRS of pairs from a population of size N. The population ratio and its estimator are

$$R = \frac{\sum_{i=1}^{N} y_i}{\sum_{i=1}^{N} x_i} = \frac{\mu_y}{\mu_x}, \qquad r = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i} = \frac{\bar{y}}{\bar{x}}.$$

Theorem 5.2: Ratio estimators

With $f = n/N$ and $s_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - r x_i)^2$,

ratio:	$r = \bar{y}/\bar{x}$,	$\widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} \cdot \dfrac{1}{\mu_x^2}\, s_r^2$	(μ_x replaced by $\bar{x}$ if unknown);
total:	$\hat{\tau}_y = r\tau_x$,	$\widehat{\mathrm{Var}}(\hat{\tau}_y) = \tau_x^2 \widehat{\mathrm{Var}}(r) = \tau_x^2 \dfrac{1-f}{n} \dfrac{s_r^2}{\bar{x}^2}$;	
mean:	$\hat{\mu}_y = r\mu_x$,	$\widehat{\mathrm{Var}}(\hat{\mu}_y) = \mu_x^2 \widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} s_r^2$.	

The residuals $y_i - r x_i$ measure the departure from a line through the origin. If $y_i \approx r x_i$ for all units, s_r^2 is small and the ratio estimator is very precise.

Derivation: first-order variance of r

Write $d_i = y_i - R x_i$. For N population units, $\bar{D} = \frac{1}{N}\sum_{i=1}^{N}(y_i - R x_i) = \mu_y - R\mu_x = 0$. If $\bar{x} \approx \mu_x$,

$$r - R = \frac{\bar{y} - R\bar{x}}{\bar{x}} \approx \frac{\bar{y} - R\bar{x}}{\mu_x} = \frac{1}{\mu_x} \cdot \frac{1}{n}\sum_{i=1}^{n}(y_i - R x_i) = \frac{\bar{d}}{\mu_x}.$$

Thus $r - R$ is approximately the SRS mean of the d_i divided by μ_x, and by Ch. 2

$$\mathrm{Var}(r) \approx \mathrm{E}(r-R)^2 = \frac{1}{\mu_x^2}\mathrm{E}(\bar{d} - \bar{D})^2 = \frac{1}{\mu_x^2} \cdot \frac{N-n}{N} \cdot \frac{1}{n} S_r^2, \qquad S_r^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - R x_i)^2.$$

Expanding the square,

$$\sum_{i=1}^{N}(y_i - R x_i)^2 = \sum_{i=1}^{N}\left[(y_i - \mu_y) - R(x_i - \mu_x)\right]^2 = (N-1)(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

using $\mu_y = R\mu_x$ and $\sum(x_i - \mu_x)(y_i - \mu_y) = (N-1)\rho S_x S_y$. Hence

$$\mathrm{Var}(r) \approx \frac{1}{\mu_x^2} \cdot \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y).$$

Replacing R by r and S_r^2 by s_r^2 gives $\widehat{\mathrm{Var}}(r)$. Note that $\mathrm{Var}(r)$ is smallest when $y_i = R x_i$ exactly.

2.1 Correlation form

Expanding s_r^2 gives the equivalent estimate

$$\widehat{\mathrm{Var}}(r) = \frac{1-f}{n} \cdot \frac{1}{\bar{x}^2}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y), \qquad \hat{\rho} = \frac{s_{xy}}{s_x s_y},$$

with $s_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$. A larger positive correlation $\hat{\rho}$ means a smaller variance.

Example 5.3: Assessed values of homes

To estimate the relative change in the assessed value of homes over two years (x: earlier value, y: current value), $n = 20$ homes give

$$r = \frac{\sum_{i=1}^{20} y_i}{\sum_{i=1}^{20} x_i} = 1.066, \qquad \sqrt{\widehat{\mathrm{Var}}(r)} = 0.00745, \qquad B = 2\sqrt{\widehat{\mathrm{Var}}(r)} = 0.0149,$$

so an approximate 95% interval for R is [1.0511, 1.0809].

Example 5.4: Oranges, continued

Ten oranges give $\sum y_i = 0.246$ (sugar) and $\sum x_i = 4.350$ (weight); the load weighs $\tau_x = 1800$ pounds:

$$\hat{\tau}_y = \frac{0.246}{4.350}(1800) = 101.79, \qquad 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau}_y)} = 6.3.$$

3 Bias of the Ratio Estimator

r is a ratio of random variables and is generally biased for R. Write $\delta_y = (\bar{y} - \mu_y)/\mu_y$ and $\delta_x = (\bar{x} - \mu_x)/\mu_x$, so

$$r = R\,(1 + \delta_y)(1 + \delta_x)^{-1} = R(1 + \delta_y)(1 - \delta_x + \delta_x^2 - \cdots) \approx R[1 + \delta_y - \delta_x + \delta_x^2 - \delta_x\delta_y].$$

- **Case 1** ($\bar{x} \approx \mu_x$, keep only δ_y): $r \approx R + R\delta_y$, so $\mathrm{E}(r) \approx R$.
- **Case 2** (first order): $\mathrm{E}(r) \approx R + R[\mathrm{E}\delta_y - \mathrm{E}\delta_x] = R$.
- **Case 3** (second order):
$$\mathrm{E}(r) - R \approx R\,\mathrm{E}(\delta_x^2) - R\,\mathrm{E}(\delta_x\delta_y).$$

Theorem 5.5: Approximate bias

$$\mathrm{E}(r) - R \approx \frac{1}{\mu_x^2} \cdot \frac{1 - f}{n}(RS_x^2 - \rho S_x S_y).$$

The bias vanishes (to this order) when $RS_x = \rho S_y$, i.e. when $R = \rho S_y/S_x = S_{xy}/S_x^2$ is the least-squares slope.

Proof. Under SRS, $\mathrm{E}[(\bar{x} - \mu_x)^2] = \frac{1-f}{n}S_x^2$ and $\mathrm{E}[(\bar{x} - \mu_x)(\bar{y} - \mu_y)] = \frac{1-f}{n}S_{xy} = \frac{1-f}{n}\rho S_x S_y$. Therefore

$$\mathrm{E}(r) - R \approx \frac{R}{\mu_x^2}\frac{1 - f}{n}S_x^2 - \frac{R}{\mu_x\mu_y}\frac{1 - f}{n}\rho S_x S_y,$$

and $R/(\mu_x\mu_y) = 1/\mu_x^2$. ∎

Remark 5.6: When is r unbiased?

If $y_i = \beta x_i$ (a line through the origin), then $R = \sum y_i / \sum x_i = \beta$ and also

$$\rho\frac{S_y}{S_x} = \frac{S_{xy}}{S_x^2} = \frac{\frac{1}{N-1}\sum(x_i - \mu_x)\,\beta(x_i - \mu_x)}{\frac{1}{N-1}\sum(x_i - \mu_x)^2} = \beta,$$

so the bias term is zero. The bias is negligible whenever the relation between y and x is a straight line through the origin.

Theorem 5.7: An exact bias bound

$$\mathrm{E}(r) - R = -\frac{\mathrm{Cov}(r, \bar{x})}{\mu_x}, \qquad \frac{|\mathrm{E}(r) - R|}{\sqrt{\mathrm{Var}(r)}} \le \frac{\sqrt{\mathrm{Var}(\bar{x})}}{\mu_x} = CV(\bar{x}).$$

Proof. Since $r\bar{x} = \bar{y}$,

$$\mathrm{Cov}(r, \bar{x}) = \mathrm{E}(r\bar{x}) - \mathrm{E}(r)\mathrm{E}(\bar{x}) = \mu_y - \mathrm{E}(r)\mu_x = \mu_x(R - \mathrm{E}(r)).$$

By $|\mathrm{Cov}(r, \bar{x})| \le \sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$, $|\mathrm{E}(r) - R| \le \frac{1}{\mu_x}\sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$. ■

The bias is small relative to the standard error whenever the coefficient of variation of $\bar{x}$ is small, which holds for moderately large n.

4 Ratio Estimation in Stratified Random Sampling

With strata $h = 1, \dots, L$ (weights $w_h = N_h/N$), there are two ways to build a ratio estimator of μ_y.

Definition 5.8

1. **Separate ratio estimator**: estimate the ratio within each stratum and combine,

$$\hat{\mu}_{yRS} = \sum_{h=1}^{L} w_h \, r_h \, \mu_{xh}, \qquad r_h = \frac{\bar{y}_h}{\bar{x}_h}.$$

   For two strata A and B: $\hat{\mu}_{yRS} = \frac{N_A}{N}\frac{\bar{y}_A}{\bar{x}_A}\mu_{xA} + \frac{N_B}{N}\frac{\bar{y}_B}{\bar{x}_B}\mu_{xB}$.

2. **Combined ratio estimator**: estimate μ_y and μ_x by the stratified means and take one ratio,

$$\hat{\mu}_{yRC} = r_C \, \mu_x, \qquad r_C = \frac{\bar{y}_{\mathrm{st}}}{\bar{x}_{\mathrm{st}}}, \quad \bar{y}_{\mathrm{st}} = \sum_h w_h \bar{y}_h, \ \bar{x}_{\mathrm{st}} = \sum_h w_h \bar{x}_h.$$

Theorem 5.9: Estimated variances

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRS}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} \cdot \frac{\sum_{i=1}^{n_h}(y_{hi} - r_h x_{hi})^2}{n_h - 1},$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRC}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} s_{rh}^2,$$

where s_{rh}^2 is the sample variance of the terms $y_{hi} - r_C x_{hi}$ in stratum h.

Derivation: which one is better?

The population versions are $\mathrm{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,S}^2$ and $\mathrm{Var}(\hat{\mu}_{yRC}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,C}^2$, with $R_h = \mu_{yh}/\mu_{xh}$, $R_C = \mu_y/\mu_x$ and

$$(N_h - 1)S_{h,S}^2 = \sum_i (y_{hi} - R_h x_{hi})^2 = (N_h - 1)(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h \rho_h S_{xh} S_{yh}),$$

$$(N_h - 1)S_{h,C}^2 = \sum_i \left[(y_{hi} - \mu_{yh}) - R_C(x_{hi} - \mu_{xh})\right]^2 = (N_h - 1)(S_{yh}^2 + R_C^2 S_{xh}^2 - 2R_C \rho_h S_{xh} S_{yh}).$$

(The combined deviation $\hat{\mu}_{yRC} - \mu_y \approx \bar{y}_{\text{st}} - R_C\bar{x}_{\text{st}} = \bar{d}_{\text{st}}$ is a stratified mean of $d_{hi} = y_{hi} - R_C x_{hi}$, whose stratum means are $\mu_{yh} - R_C\mu_{xh}$.) Subtracting,

$$\begin{aligned} S^2_{h,C} - S^2_{h,S} &= (R_C^2 - R_h^2)S^2_{xh} - 2(R_C - R_h)\rho_h S_{xh}S_{yh} \\ &= (R_C - R_h)^2 S^2_{xh} + 2(R_C - R_h)(R_h S^2_{xh} - \rho_h S_{xh}S_{yh}). \end{aligned}$$

The last bracket is zero when each stratum ratio is (approximately) unbiased, $R_h S_{xh} = \rho_h S_{yh}$. Then

$$\text{Var}(\hat{\mu}_{yRC}) - \text{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h}(R_C - R_h)^2 S^2_{xh} \ \geq 0.$$

Remark 5.10: Practical guidance

- When the stratum ratios R_h differ greatly, the separate estimator is more efficient.
- But each r_h carries its own ratio bias, and these add up across strata. When the n_h are small the separate estimator can be badly biased; when the n_h are large (over about 50) the bias is negligible. With small strata the combined estimator is safer.

5 Ratio Estimator versus the Sample Mean

Compare $\bar{y}$ and $\hat{\mu}_y = r\mu_x$ by relative efficiency $\text{RE}(\hat{\theta}_1, \hat{\theta}_2) = \text{Var}(\hat{\theta}_1)/\text{Var}(\hat{\theta}_2)$:

$$\text{Var}(\bar{y}) = \frac{1-f}{n}S_y^2, \qquad \text{Var}(\hat{\mu}_y) \approx \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

$$\text{RE}(\bar{y}, \hat{\mu}_y) = \frac{\text{Var}(\bar{y})}{\text{Var}(\hat{\mu}_y)} = \frac{S_y^2}{S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y}.$$

Theorem 5.11: When the ratio estimator wins

$\text{RE}(\bar{y}, \hat{\mu}_y) > 1$ if and only if

$$\rho > \frac{1}{2} \cdot \frac{S_x/\mu_x}{S_y/\mu_y} = \frac{1}{2} \cdot \frac{CV_x}{CV_y}.$$

In particular, if x and y have equal coefficients of variation, the ratio estimator is better as soon as $\rho > 1/2$.

Proof. $\text{RE} > 1 \iff R^2 S_x^2 < 2R\rho S_x S_y \iff RS_x < 2\rho S_y \iff \rho > \frac{RS_x}{2S_y}$, and $R = \mu_y/\mu_x$. The sample version replaces everything by estimates: $\hat{\rho} > \frac{1}{2}\frac{s_x/\bar{x}}{s_y/\bar{y}}$. ■

The ratio estimator is more efficient than $\bar{y}$ when x and y are highly positively correlated; the estimated relative efficiency is meaningful only for large samples.

Chapter 6. Simple Linear Regression

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This chapter reviews simple linear regression, the tool behind the regression estimator of Ch. 7. Regression compresses n-dimensional data into a 2-dimensional summary $(\hat{\beta}_0, \hat{\beta}_1)$: data = model structure + error.

1 The Model

Definition 6.1

The simple linear regression model with a single regressor x is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where the errors are uncorrelated with mean 0 and unknown variance σ^2. The regressor x is treated as fixed; y is random. β_0 (intercept) and β_1 (slope) are the **regression coefficients**.

Then

$$\mathrm{E}(y \mid x) = \beta_0 + \beta_1 x = f(x), \qquad \mathrm{Var}(y \mid x) = \mathrm{Var}(\varepsilon) = \sigma^2,$$

so the true regression line is the line of mean responses. The goals are to estimate β_0, β_1 and to test $H_0 : \beta_1 = 0$.

2 Least-Squares Estimation

The least-squares (LS) estimators minimize

$$S(\beta_0,\beta_1)=\sum_{i=1}^{n}(y_i-\beta_0-\beta_1x_i)^2=\sum_{i=1}^{n}\varepsilon_i^2.$$

Setting $\partial S/\partial\beta_0=\partial S/\partial\beta_1=0$ gives the **normal equations**

$$n\hat\beta_0+\hat\beta_1\sum x_i=\sum y_i,\qquad \hat\beta_0\sum x_i+\hat\beta_1\sum x_i^2=\sum x_iy_i.$$

Theorem 6.2: LS estimators

$$\hat\beta_1=\frac{S_{xy}}{S_{xx}}=\frac{\sum(x_i-\bar x)(y_i-\bar y)}{\sum(x_i-\bar x)^2}=\frac{\sum(x_i-\bar x)y_i}{\sum(x_i-\bar x)^2},\qquad \hat\beta_0=\bar y-\hat\beta_1\bar x.$$

The **fitted values** are $\hat y_i=\hat\beta_0+\hat\beta_1x_i=\widehat{\mathrm{E}}(y_i\mid x_i)$ and the **residuals** are $e_i=y_i-\hat y_i$, so $y_i=\hat\beta_0+\hat\beta_1x_i+e_i$; residuals are the basis of model-adequacy checks (residual analysis).

Example 6.3: Rocket propellant data, $n=20$

$y=$ shear strength, $x=$ age of propellant. From $\bar x=13.3625$, $\bar y=2131.358$, $S_{xx}=1106.559$, $S_{xy}=-41112.65$:

$$\hat\beta_1=\frac{-41112.65}{1106.559}=-37.15,\qquad \hat\beta_0=2131.358+37.15\times13.3625=2627.82,$$

so $\hat y=2627.82-37.15x$. R output: intercept 2627.822 (SE 44.184), slope -37.154 (SE 2.889, $t=-12.86$, $p=1.64\times10^{-10}$).

2.1 Properties of the estimators

Theorem 6.4

$\mathrm{E}(\hat\beta_1)=\beta_1$, $\mathrm{E}(\hat\beta_0)=\beta_0$, and

$$\mathrm{Var}(\hat\beta_1)=\frac{\sigma^2}{S_{xx}},\qquad \mathrm{Var}(\hat\beta_0)=\sigma^2\Big(\frac1n+\frac{\bar x^2}{S_{xx}}\Big),\qquad \mathrm{Cov}(\hat\beta_0,\hat\beta_1)=-\bar x\frac{\sigma^2}{S_{xx}}.$$

Derivation: proof

Write $\hat\beta_1=\sum c_iy_i$ with $c_i=(x_i-\bar x)/S_{xx}$. Then $\sum c_i=0$ and $\sum c_ix_i=\sum(x_i-\bar x)x_i/S_{xx}=1$, so

$$\mathrm{E}(\hat\beta_1)=\sum c_i(\beta_0+\beta_1x_i)=\beta_0\sum c_i+\beta_1\sum c_ix_i=\beta_1,$$

$$\mathrm{Var}(\hat\beta_1)=\sigma^2\sum c_i^2=\sigma^2\frac{S_{xx}}{S_{xx}^2}=\frac{\sigma^2}{S_{xx}}.$$

Since $\mathrm{E}(\bar y)=\beta_0+\beta_1\bar x$, $\mathrm{E}(\hat\beta_0)=\beta_0+\beta_1\bar x-\bar x\beta_1=\beta_0$. With $\bar y=\sum d_iy_i$, $d_i=1/n$,

$$\mathrm{Cov}(\bar y,\hat\beta_1)=\sigma^2\sum d_ic_i=\frac{\sigma^2}{n}\sum c_i=0,$$

hence

$$\mathrm{Var}(\hat\beta_0)=\mathrm{Var}(\bar y)+\bar x^2\mathrm{Var}(\hat\beta_1)-2\bar x\mathrm{Cov}(\bar y,\hat\beta_1)=\sigma^2\Big(\frac1n+\frac{\bar x^2}{S_{xx}}\Big),$$

$$\mathrm{Cov}(\hat\beta_0,\hat\beta_1)=\mathrm{Cov}(\bar y,\hat\beta_1)-\bar x\mathrm{Var}(\hat\beta_1)=-\bar x\sigma^2/S_{xx}.$$

2.2 Properties of the fit

Theorem 6.5

(1) $\sum e_i=0$, equivalently $\sum y_i=\sum\hat y_i$;

(2) the fitted line passes through $(\bar x,\bar y)$;

(3) $\sum x_ie_i=0$;

(4) $\sum\hat y_ie_i=0$.

Proof. (1) and (3) are the two normal equations: $\sum(y_i-\hat\beta_0-\hat\beta_1x_i)=0$ and $\sum x_i(y_i-\hat\beta_0-\hat\beta_1x_i)=0$. (2): at $x=\bar x$, $\hat y=\hat\beta_0+\hat\beta_1\bar x=\bar y$. (4): $\sum\hat y_ie_i=\hat\beta_0\sum e_i+\hat\beta_1\sum x_ie_i=0$. ■

2.3 Centered model

Rewriting

$$y_i=(\beta_0+\beta_1\bar x)+\beta_1(x_i-\bar x)+\varepsilon_i=\beta_0'+\beta_1(x_i-\bar x)+\varepsilon_i$$

gives $\hat\beta_0'=\bar y$ and $\hat\beta_1=S_{xy}/S_{xx}$, and now $\mathrm{Cov}(\hat\beta_0',\hat\beta_1)=\mathrm{Cov}(\bar y,\hat\beta_1)=0$: centering makes the estimators uncorrelated.

3 Estimation of σ^2

Theorem 6.6: Sum-of-squares decomposition

$$\underbrace{\sum(y_i-\bar y)^2}_{SS_T}=\underbrace{\sum(y_i-\hat y_i)^2}_{SS_E}+\underbrace{\sum(\hat y_i-\bar y)^2}_{SS_R},\qquad SS_E=SS_T-\hat\beta_1S_{xy},\quad SS_R=\hat\beta_1S_{xy}.$$

Proof. Expanding $\sum(y_i-\hat y_i+\hat y_i-\bar y)^2$ leaves the cross term $2\sum e_i(\hat y_i-\bar y)=2(\sum e_i\hat y_i-\bar y\sum e_i)=0$. Since $e_i=(y_i-\bar y)-\hat\beta_1(x_i-\bar x)$,

$$SS_E=SS_T+\hat\beta_1^2S_{xx}-2\hat\beta_1S_{xy}=SS_T-\hat\beta_1S_{xy}.$$

■

SS_E has $n-2$ degrees of freedom, and the unbiased estimator of σ^2 is

$$\hat\sigma^2=\frac{SS_E}{n-2}=MS_E\quad\text{(the mean square error)}.$$

$\hat\sigma$ is the standard error of regression; it is a model-dependent estimate of σ.

Example 6.7: Rocket propellant

$SS_T=1{,}693{,}737.6$ and $SS_E=1{,}693{,}737.6-(-37.15)(-41112.65)=166{,}402.65$, so $\hat\sigma^2=166402.65/18=9244.59$.

4 Hypothesis Testing on the Slope

Now assume $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Then $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

Theorem 6.8: t-test for significance of regression

For $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$,

$$t_0 = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)}, \qquad \text{se}(\hat{\beta}_1) = \sqrt{MS_E/S_{xx}},$$

which is t_{n-2} under H_0. Reject H_0 if $|t_0| > t_{\alpha/2,n-2}$ (equivalently, if the P-value $\leq \alpha$). If σ^2 were known, $Z_0 = \hat{\beta}_1/\sqrt{\sigma^2/S_{xx}} \sim N(0,1)$ would be used instead.

Failing to reject H_0 means there is no evidence of a *linear* relationship between x and y.

Example 6.9: Rocket propellant

$t_0 = -37.15/\sqrt{9244.59/1106.56} = -12.85$ and $t_{0.025,18} = 2.101$, so H_0 is rejected; the P-value is 1.64×10^{-10}.

4.1 Analysis of variance

Theorem 6.10: F-test

With degrees of freedom $(n-1) = 1 + (n-2)$,

$$F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{MS_R}{MS_E} \sim F_{1,n-2} \quad \text{under } H_0,$$

and H_0 is rejected if $F_0 > F_{\alpha,1,n-2}$. Since $\text{E}(MS_R) = \sigma^2 + \beta_1^2 S_{xx}$ and $\text{E}(MS_E) = \sigma^2$, F_0 is near 1 under H_0 and large otherwise. In simple regression $F_0 = t_0^2$.

ANOVA gives the same answer as the t-test here, but it is the natural tool for multiple regression.

Example 6.11: Rocket propellant

$SS_R = (-37.15)(-41112.65) = 1{,}527{,}334.95$, $SS_E = 166{,}402.65$,

$$F_0 = \frac{1527334.95/1}{166402.65/18} = 165.21 > F_{0.05,1,18} = 4.41,$$

so H_0 is rejected (R: $F = 165.4$ on 1 and 18 df, $p = 1.64 \times 10^{-10}$).

5 Interval Estimation

5.1 Coefficients

Under normal errors, $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j) \sim t_{n-2}$, so

$$\hat{\beta}_1 \pm t_{\alpha/2,n-2}\,\text{se}(\hat{\beta}_1), \qquad \hat{\beta}_0 \pm t_{\alpha/2,n-2}\,\text{se}(\hat{\beta}_0),$$

$$\text{se}(\hat\beta_1) = \sqrt{\frac{MS_E}{S_{xx}}}, \qquad \text{se}(\hat\beta_0) = \sqrt{MS_E\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

Example 6.12

For the rocket data with $t_{0.025,18} = 2.101$:

$$2534.99 \le \beta_0 \le 2720.65, \qquad -43.22 \le \beta_1 \le -31.08.$$

5.2 Error variance

Since $(n-2)MS_E/\sigma^2 \sim \chi^2_{n-2}$,

$$\frac{(n-2)MS_E}{\chi^2_{\alpha/2,n-2}} \le \sigma^2 \le \frac{(n-2)MS_E}{\chi^2_{1-\alpha/2,n-2}}.$$

5.3 Mean response

For x_0 within the range of the data, the estimate is $\hat\mu_{y|x_0} = \hat\beta_0 + \hat\beta_1 x_0$, with $\text{E}(\hat\mu_{y|x_0}) = \beta_0 + \beta_1 x_0$ and

$$\begin{aligned}\text{Var}(\hat\mu_{y|x_0}) &= \text{Var}(\hat\beta_0) + x_0^2\text{Var}(\hat\beta_1) + 2x_0\text{Cov}(\hat\beta_0,\hat\beta_1)\\ &= \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big) + x_0^2\frac{\sigma^2}{S_{xx}} - 2x_0\bar x\frac{\sigma^2}{S_{xx}} = \sigma^2\Big(\frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big).\end{aligned}$$

Hence

$$\hat\mu_{y|x_0} \pm t_{\alpha/2,n-2}\sqrt{MS_E\Big(\frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big)}.$$

At $x_0 = \bar x = 13.3625$ in the rocket data the 95% interval is $2086.2 \le \text{E}(y \mid x_0) \le 2176.6$. The interval is narrowest at $\bar x$ and widens away from it.

5.4 Prediction of a new observation

The point prediction of y_0 at x_0 is $\hat y_0 = \hat\beta_0 + \hat\beta_1 x_0$. Since the new y_0 is independent of the data,

$$\text{E}(\hat y_0 - y_0)^2 = \text{E}[\hat y_0 - \text{E}(y_0)]^2 + \text{E}[y_0 - \text{E}(y_0)]^2 = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big),$$

and the $100(1-\alpha)\%$ prediction interval is

$$\hat y_0 \pm t_{\alpha/2,n-2}\sqrt{MS_E\Big(1 + \frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big)}.$$

For the rocket data at $x_0 = 10$: $[2048.32,\ 2464.32]$.

6 Coefficient of Determination

Definition 6.13

$R^2 = \dfrac{SS_R}{SS_T} = 1 - \dfrac{SS_E}{SS_T}$, $0 \le R^2 \le 1$: the proportion of variation in y explained by x.

In simple regression,

$$R^2 = \frac{\hat{\beta}_1 S_{xy}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r_{xy}^2, \qquad F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{R^2(n-2)}{1-R^2}.$$

For the rocket data $R^2 = 1527334.95/1693737.6 = 0.9018 = (-0.9497)^2$: 90.18% of the variability in strength is explained by age.

Chapter 7. Regression Estimation

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1 The Regression Estimator

The ratio estimator works best when y is proportional to x (a line through the origin). If the relation is linear with a nonzero intercept, use the regression line instead. As in Ch. 6 the x's are treated as fixed (already observed) and the y's as random:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \beta_0' + \beta_1(x_i - \bar{x}) + \varepsilon_i, \qquad \hat{\beta}_0' = \bar{y}, \quad \hat{\beta}_1 = b = \frac{\sum(y_i - \bar{y})(x_i - \bar{x})}{\sum(x_i - \bar{x})^2},$$

so the fitted mean response is $\widehat{\mathrm{E}}(y \mid x) = \bar{y} + b(x - \bar{x})$. Evaluating it at the known population mean μ_x gives the estimator.

Definition 7.1

The **regression estimator** of μ_y is

$$\hat{\mu}_{yL} = \widehat{\mathrm{E}}(y \mid \mu_x) = \bar{y} + b(\mu_x - \bar{x}) = \frac{1}{n}\sum_{i=1}^{n}\left[y_i - b(x_i - \mu_x)\right],$$

i.e. the sample mean of the adjusted values $y_i - b(x_i - \mu_x)$.

If the sample happens to have $\bar{x} < \mu_x$ and $b > 0$, the sample mean $\bar{y}$ is probably too small as well, and $\hat{\mu}_{yL}$ corrects it upward by $b(\mu_x - \bar{x})$.

Theorem 7.2: Model unbiasedness

Under the model $\mathrm{E}(y_i) = \beta_0 + \beta_1 x_i$ with fixed x's,

$$\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \mu_x = \mathrm{E}(y \mid \mu_x).$$

Proof. $\mathrm{E}(\bar{y}) = \beta_0 + \beta_1 \bar{x}$ and $\mathrm{E}(b) = \beta_1$ (Ch. 6), so $\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \bar{x} + \beta_1(\mu_x - \bar{x}) = \beta_0 + \beta_1 \mu_x$. ■

2 Variance of the Regression Estimator

2.1 Fixed slope

First take a fixed constant β and $\bar{y}_{\text{reg}} = \bar{y} + \beta(\mu_x - \bar{x})$. This is the SRS mean of $u_i = y_i - \beta(x_i - \mu_x)$, whose population mean is μ_y, so it is exactly unbiased.

Theorem 7.3

$$\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n}(S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2),$$

which is minimized at $\beta = S_{xy}/S_x^2$ (the population LS slope), where

$$\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n}\Big(S_y^2 - \frac{S_{xy}^2}{S_x^2}\Big) = \frac{1-f}{n} S_y^2(1-\rho^2).$$

Proof. By Ch. 2, $\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n} S_u^2$ with

$$\begin{aligned} S_u^2 &= \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i-\mu_y) - \beta(x_i-\mu_x)\big]^2 \\ &= \frac{1}{N-1}\Big[\sum(y_i-\mu_y)^2 - 2\beta\sum(x_i-\mu_x)(y_i-\mu_y) + \beta^2\sum(x_i-\mu_x)^2\Big] = S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2. \end{aligned}$$

This quadratic in β is minimized at $\beta = S_{xy}/S_x^2$, giving $S_y^2 - S_{xy}^2/S_x^2 = S_y^2(1-\rho^2)$. ∎

2.2 Estimated slope

With b estimated, write $\xi_i = (y_i - \mu_y) - \beta(x_i - \mu_x)$ for $\beta = S_{xy}/S_x^2$, so $y_i = \mu_y + \beta(x_i - \mu_x) + \xi_i$. Then

$$b = \frac{\sum(x_i-\bar{x})y_i}{\sum(x_i-\bar{x})^2} = \beta\frac{\sum(x_i-\bar{x})(x_i-\mu_x)}{\sum(x_i-\bar{x})^2} + \frac{\sum(x_i-\bar{x})\xi_i}{\sum(x_i-\bar{x})^2} = \beta + \frac{\sum(x_i-\bar{x})\xi_i}{\sum(x_i-\bar{x})^2},$$

and the last term is of smaller order since the ξ_i are uncorrelated with x in the population. Hence $b \approx \beta$ in large samples and

$$\operatorname{Var}(\hat{\mu}_{yL}) \approx \operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n} S^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i-\mu_y) - \beta(x_i-\mu_x)\big]^2.$$

Theorem 7.4: Estimated variance

$$\widehat{\operatorname{Var}}(\hat{\mu}_{yL}) = \frac{N-n}{Nn} MS_E, \qquad MS_E = \frac{1}{n-2}\sum_{i=1}^{n}\big[(y_i-\bar{y}) - b(x_i-\bar{x})\big]^2,$$

the mean square error from the simple linear regression of y on x.

The sample analogue of S^2 is $\frac{1}{n-1}\sum e_i^2$ with residuals $e_i = (y_i - \bar{y}) - b(x_i - \bar{x})$; dividing by $n-2$ instead (two estimated coefficients) gives MS_E, and the difference is negligible for moderate n.

Example 7.5: Calculus marks

A mathematics achievement test was taken by all $N = 486$ entering students, with mean $\mu_x = 52$. For $n = 10$ sampled students:

Student	1	2	3	4	5	6	7	8	9	10
Test (x)	39	43	21	64	57	47	28	75	34	52
Final (y)	65	78	52	82	92	89	73	98	56	75

Here $\bar{x} = 46$, $\bar{y} = 76$, and the R fit gives $\hat{y} = 40.78 + 0.7656x$ with residual standard error 8.704 on 8 df ($MS_E = 75.76$, $R^2 = 0.705$). Then

$$\hat{\mu}_{yL} = 76 + 0.7656(52 - 46) = 80.6,$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{486 - 10}{486 \times 10}(75.76) = 7.42, \qquad B = 2\sqrt{7.42} = 5.45.$$

The fitted intercept (40.8) is far from zero, so a ratio estimator would be inappropriate here: $r\mu_x = (760/460)(52) = 85.9$ overshoots.

3 Relative Efficiency: Mean, Ratio and Regression

For two estimators E_1, E_2 of μ_y, $\mathrm{RE}(E_1/E_2) = \mathrm{Var}(E_2)/\mathrm{Var}(E_1)$; if it exceeds 1, E_1 is preferred. The estimated version (with estimated variances) is meaningful for large samples. The three estimated variances share the factor $\frac{1-f}{n}$:

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{1-f}{n}s_y^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_y) = \frac{1-f}{n}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y), \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{1-f}{n}s_y^2(1-\hat{\rho}^2).$$

Derivation: the regression variance in correlation form

$$\begin{aligned}\frac{1}{n-1}\sum\left[(y_i - \bar{y}) - b(x_i - \bar{x})\right]^2 &= \frac{1}{n-1}\left[S_{yy} - 2b\,S_{xy} + b^2 S_{xx}\right] \\ &= s_y^2 - 2b\,s_{xy} + b^2 s_x^2 = s_y^2 - b^2 s_x^2,\end{aligned}$$

since $s_{xy} = b s_x^2$. Moreover $\hat{\rho} = \frac{s_{xy}}{s_x s_y} = \frac{b s_x^2}{s_x s_y}$ gives $\hat{\rho} s_y = b s_x$, so $s_y^2 - b^2 s_x^2 = s_y^2(1-\hat{\rho}^2)$.

3.1 Mean versus ratio

From Ch. 5,

$$\widehat{\mathrm{RE}}(\hat{\mu}_y/\bar{y}) = \frac{s_y^2}{s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y},$$

and the ratio estimator is more efficient iff $\hat{\rho} > \frac{1}{2}\cdot\frac{s_x/\bar{x}}{s_y/\bar{y}}$. Recall that r is generally biased for R; the bias is negligible when the relation is a line through the origin.

3.2 Mean versus regression

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\bar{y}) = \frac{s_y^2}{s_y^2(1-\hat{\rho}^2)} = \frac{1}{1-\hat{\rho}^2} > 1 \quad \text{unless } \hat{\rho} = 0.$$

The regression estimator is (asymptotically) never worse than $\bar{y}$.

3.3 Ratio versus regression

Theorem 7.6

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\hat{\mu}_y) = \frac{s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y}{s_y^2(1-\hat{\rho}^2)} \geq 1,$$

with equality iff $r = b$.

Proof. The numerator minus the denominator is

$$r^2 s_x^2 - 2r\hat{\rho}s_x s_y + \hat{\rho}^2 s_y^2 = (rs_x - \hat{\rho}s_y)^2 = (rs_x - bs_x)^2 = s_x^2(r-b)^2 \geq 0.$$

■

Both r and b are linear in the y's: $r = \sum y_i / \sum x_i = \sum d_i y_i$ with equal weights $d_i = 1/\sum x_j$, while $b = \sum w_i y_i$ with $w_i = (x_i - \bar{x})/\sum (x_j - \bar{x})^2$. They coincide only when the LS line passes through the origin, which is exactly when the ratio estimator is as good as the regression estimator.

Chapter 8. Cluster Sampling

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1 Introduction

Suppose we want the weekly food expenditure of families in Seoul. SRS, stratified or systematic sampling of families requires a list of all families (rarely available), and the sampled families would be scattered all over the city, which makes interviewing expensive. A list of city blocks, however, is easy to obtain, and interviewing every family in a few blocks is cheap.

Definition 8.1

A **cluster sample** is a simple random sample in which each sampling unit is a collection (cluster) of elements; every element of a selected cluster is observed.

Cluster sampling is effective when

- a good frame of elements is unavailable or costly, while a frame of clusters is easy to obtain;
- the cost of observation increases with the distance between elements.

Since it is SRS of clusters, the estimators parallel those of Ch. 2, with cluster totals playing the role of observations.

Notation.

N	number of clusters in the population
n	number of clusters selected by SRS
m_i	number of elements in cluster i, $i = 1, \ldots, N$
$\bar{m} = \frac{1}{n}\sum_{i=1}^{n} m_i$	average cluster size in the sample
$M = \sum_{i=1}^{N} m_i$	number of elements in the population
$\bar{M} = M/N$	average cluster size in the population
y_{ij}	value of the jth element in cluster i
$y_i = \sum_{j=1}^{m_i} y_{ij}$	total of cluster i

2 Estimation of the Population Mean

The population mean per element is

$$\mu = \frac{\sum_{i=1}^{N}\sum_{j=1}^{m_i} y_{ij}}{\sum_{i=1}^{N} m_i} = \frac{\sum_{i=1}^{N} y_i}{M},$$

a ratio of the cluster totals y_i to the cluster sizes m_i. Its natural estimator is therefore a ratio estimator (Ch. 5) with $x_i = m_i$.

Theorem 8.2: Cluster estimator of μ

$$\bar{y}_{\text{cl}} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}\, s_c^2, \qquad s_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_{\text{cl}} m_i)^2,$$

where $\bar{M}$ is replaced by $\bar{m}$ if M is unknown. An approximate 95% interval is $\bar{y}_{\text{cl}} \pm 2\sqrt{\widehat{\text{Var}}(\bar{y}_{\text{cl}})}$.

Derivation: from the ratio estimator

With $x_i = m_i$, $R = \mu$ and $\mu_x = \bar{M}$, Ch. 5 gives

$$\text{Var}(\bar{y}_{\text{cl}}) \approx \frac{1}{\bar{M}^2}\cdot\frac{N-n}{N}\cdot\frac{1}{n}\cdot\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \mu m_i)^2 = \frac{1}{\bar{M}^2}\frac{N-n}{Nn}(S_y^2 + \mu^2 S_m^2 - 2\mu\rho S_m S_y),$$

and replacing population quantities by sample ones yields $\widehat{\text{Var}}(\bar{y}_{\text{cl}})$ and the alternative form

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}(s_y^2 + \bar{y}_{\text{cl}}^2 s_m^2 - 2\bar{y}_{\text{cl}}\hat{\rho} s_m s_y), \qquad s_m^2 = \frac{1}{n-1}\sum_{i=1}^{n}(m_i - \bar{m})^2,$$

where s_y^2 is the sample variance of the cluster totals. Cluster totals nearly proportional to cluster sizes ($y_i \approx \mu m_i$) give a small variance.

2.1 A regression-type estimator

Treating the cluster size as the covariate, $y_i = \beta_0 + \beta_1 m_i + \varepsilon_i = \beta_0' + \beta_1(m_i - \bar{m}) + \varepsilon_i$, the regression estimator of the mean cluster total is

$$\hat{\mu}_t = \bar{y}_t + \hat{\beta}_1(\bar{M} - \bar{m}), \qquad \hat{\beta}_1 = \frac{\sum(m_i - \bar{m})(y_i - \bar{y}_t)}{\sum(m_i - \bar{m})^2}, \quad \bar{y}_t = \frac{1}{n}\sum y_i,$$

with $\widehat{\text{Var}}(\hat{\mu}_t) = \frac{N-n}{Nn} MS_E$ as in Ch. 7. Dividing by $\bar{M}$ gives an estimator of μ. It beats the ratio form $\bar{y}_{\text{cl}}$ unless the LS line of y_i on m_i passes through the origin.

3 Estimation of the Population Total

3.1 M known

Since $\tau = M\mu$,

$$\hat{\tau} = M\bar{y}_{\text{cl}} = M\frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\hat{\tau}) = M^2\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = N^2\frac{N-n}{Nn}s_c^2.$$

3.2 M unknown

Often the number of elements is unknown. Then use the mean of the cluster totals,

$$\bar{y}_t = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \text{E}(\bar{y}_t) = \frac{1}{N}\sum_{i=1}^{N} y_i = \frac{\tau}{N},$$

which is unbiased for the average cluster total, so $N\bar{y}_t$ is unbiased for τ:

$$\hat{\tau} = N\bar{y}_t = \frac{N}{n}\sum_{i=1}^{n} y_i, \qquad \widehat{\text{Var}}(\hat{\tau}) = N^2\frac{N-n}{Nn}s_t^2, \qquad s_t^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_t)^2.$$

This is SRS of the cluster totals and ignores the sizes m_i; it is less precise than $M\bar{y}_{\text{cl}}$ when the totals vary mainly because the sizes vary.

If all $m_i = m$, then $M = Nm$, $\bar{y}_{\text{cl}} = \bar{y}_t/m$, and the two estimators of τ coincide.

4 Comparison with Simple Random Sampling

Assume equal cluster sizes $m_i = m$, so $M = Nm$ and

$$\bar{y}_{\text{cl}} = \frac{\sum_{i=1}^{n} y_i}{nm} = \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} y_{ij} = \frac{\bar{y}_t}{m}.$$

Let $\bar{y}_i = y_i/m$ be the mean of cluster i. Then

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{1}{m^2}\widehat{\text{Var}}(\bar{y}_t) = \frac{1}{m^2}\frac{N-n}{Nn}\frac{1}{n-1}\sum_i (m\bar{y}_i - m\bar{y}_{\text{cl}})^2 = \frac{N-n}{Nn}\frac{1}{n-1}\sum_{i=1}^{n}(\bar{y}_i - \bar{y}_{\text{cl}})^2.$$

Treat the nm observed elements as an $m \times n$ table with column means $\bar{y}_i$ and grand mean $\bar{y}_{\text{cl}}$:

$$\underbrace{\sum_{i=1}^{n}\sum_{j=1}^{m}(y_{ij} - \bar{y}_{\text{cl}})^2}_{SST} = \underbrace{\sum_i\sum_j (y_{ij} - \bar{y}_i)^2}_{SSW} + \underbrace{m\sum_i (\bar{y}_i - \bar{y}_{\text{cl}})^2}_{SSB},$$

with $MSB = SSB/(n-1)$ and $MSW = SSW/[n(m-1)]$. Then

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nnm}\,MSB.$$

If the same nm elements had been drawn by SRS from the Nm elements, with sample variance s^2,

$$\widehat{\text{Var}}(\bar{y}) = \frac{Nm - nm}{Nm}\cdot\frac{s^2}{nm} = \frac{N-n}{Nnm}\,s^2.$$

Remark 8.3: Efficiency of cluster sampling

The relative efficiency of cluster sampling versus SRS is measured by comparing MSB with s^2. Cluster sampling is more efficient when there is little variation between clusters, i.e. when each cluster is a miniature of the whole population (heterogeneous within). In most real populations, neighbouring elements are alike, $MSB > s^2$, and cluster sampling is less efficient per element than SRS; it is chosen for its lower cost per element.

5 Selecting the Sample Size

Setting $B = 2\sqrt{\frac{N-n}{NnM^2}s_c^2}$ and solving as in Ch. 2:

- for μ with $\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND + s_c^2}$, $D = \dfrac{B^2\bar{M}^2}{4}$;
- for τ with $M\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND + s_c^2}$, $D = \dfrac{B^2}{4N^2}$;
- for τ with $N\bar{y}_t$: $n = \dfrac{Ns_t^2}{ND + s_t^2}$, $D = \dfrac{B^2}{4N^2}$,

with s_c^2, s_t^2 taken from a previous survey or a pilot sample.

6 Estimation of a Population Proportion

Let a_i be the number of elements in cluster i having the characteristic. Then p is the ratio $\sum a_i / \sum m_i$ over the population, and

$$\hat{p}_{\text{cl}} = \frac{\sum_{i=1}^n a_i}{\sum_{i=1}^n m_i}, \qquad \widehat{\text{Var}}(\hat{p}_{\text{cl}}) = \frac{N-n}{NnM^2}s_p^2, \qquad s_p^2 = \frac{1}{n-1}\sum_{i=1}^n (a_i - \hat{p}_{\text{cl}}m_i)^2,$$

with $\bar{M} \approx \bar{m}$ if M is unknown. The sample size for a bound B is $n = Ns_p^2/(ND + s_p^2)$ with $D = B^2\bar{M}^2/4$.

7 Cluster Sampling Combined with Stratification

To survey the number of books read by 4th and 5th graders, divide the population into two strata (grades) and take a cluster sample (of classes) in each. In stratum ℓ let N_ℓ be the number of clusters, $\bar{y}_{t\ell}$ the mean cluster total and $\bar{m}_\ell$ the mean cluster size in the sample. Then

$$\text{average cluster total: } \frac{1}{N}(N_1\bar{y}_{t1} + N_2\bar{y}_{t2}), \qquad \text{average cluster size: } \frac{1}{N}(N_1\bar{m}_1 + N_2\bar{m}_2),$$

$$\bar{y}_c = \frac{N_1\bar{y}_{t1} + N_2\bar{y}_{t2}}{N_1\bar{m}_1 + N_2\bar{m}_2} = \frac{\sum_\ell N_\ell\bar{y}_{t\ell}}{\sum_\ell N_\ell\bar{m}_\ell},$$

a combined ratio estimator. Its estimated variance is

$$\widehat{\text{Var}}(\bar{y}_c) = \frac{1}{\left(\sum_\ell N_\ell\bar{m}_\ell\right)^2}\sum_{\ell=1}^L N_\ell(N_\ell - n_\ell)\frac{s_{t\ell}^2}{n_\ell},$$

where $s_{t\ell}^2 = \frac{1}{n_\ell - 1}\sum_{i=1}^{n_\ell}(y_i - \bar{y}_{t\ell})^2$ is the sample variance of the cluster totals in stratum ℓ. The numerator is $\sum_\ell N_\ell^2\widehat{\text{Var}}(\bar{y}_{t\ell})$, the estimated variance of the stratified estimate of the population total, and the denominator is the squared estimated number of elements.

8 Cluster Sampling with Probabilities Proportional to Size

8.1 The PPS estimator

Let the population consist of units with values $y_1, \dots, y_N$ and suppose sampling is with replacement, with z_i the probability that unit i is selected on any one draw ($\sum z_i = 1$).

Theorem 8.4: Hansen–Hurwitz estimator

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{z_i}$$

is unbiased for τ for any choice of $z_i > 0$, with $\widehat{\text{Var}}(\hat{\tau}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}\left(\frac{y_i}{z_i} - \hat{\tau}_{\text{pps}}\right)^2$.

Proof. Each draw gives an iid value $t = y_I/z_I$ with $P(I = k) = z_k$, so $\text{E}(t) = \sum_k z_k \frac{y_k}{z_k} = \tau$. The average of n iid unbiased values is unbiased, and its variance is estimated by s_t^2/n. ■

With $z_i = 1/N$ (equal probability), $\hat{\tau}_{\text{pps}} = \frac{N}{n}\sum y_i = N\bar{y}$, the usual estimator. The variance is small when $z_i \propto y_i$, so we want z_i roughly proportional to y_i.

8.2 Application to clusters

Cluster totals tend to be proportional to cluster sizes, so choose $z_i = m_i/M$. With y_i the cluster total and $\bar{y}_i = y_i/m_i$ the cluster mean,

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{m_i/M} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\text{pps}} = \frac{\hat{\tau}_{\text{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i :$$

the estimator of μ is the plain average of the sampled cluster means.

Theorem 8.5

$$\widehat{\text{Var}}(\hat{\tau}_{\text{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\text{pps}})^2, \qquad \widehat{\text{Var}}(\hat{\mu}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\text{pps}})^2.$$

Proof. The $\bar{y}_i$ are iid draws (with replacement), so $\widehat{\text{Var}}(\frac{1}{n}\sum\bar{y}_i) = \frac{1}{n}\cdot\frac{1}{n-1}\sum(\bar{y}_i - \hat{\mu}_{\text{pps}})^2$; multiply by M^2 for the total. ■

Remark 8.6: Equal sizes

If $m_i = m$, then $\hat{\mu}_{\text{pps}} = \frac{1}{n}\sum_i\frac{1}{m}\sum_j y_{ij} = \bar{y}_{\text{cl}} = \bar{y}_t/m$, and since $s_t^2 = \frac{m^2}{n-1}\sum(\bar{y}_i - \bar{y}_{\text{cl}})^2$,

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{1}{m^2}\frac{N-n}{Nn}s_t^2 = \frac{N-n}{N}\cdot\frac{1}{n(n-1)}\sum(\bar{y}_i - \bar{y}_{\text{cl}})^2,$$

which is $\widehat{\text{Var}}(\hat{\mu}_{\text{pps}})$ up to the fpc (PPS draws are with replacement).

Chapter 9. Two-Stage Cluster Sampling

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1 Introduction

Definition 9.1

A **two-stage cluster sample** is obtained by selecting a probability sample of clusters and then a probability sample of elements from each selected cluster. The clusters are the **primary sampling units** (psu, e.g. classes) and the elements the **secondary sampling units** (ssu, e.g. students).

Measuring every element in a cluster is wasteful when the elements of a cluster are similar; subsampling lets us spread the same budget over more clusters.

Notation. Both stages use SRS without replacement.

N, n	number of clusters in the population / selected
M_i, m_i	number of elements in cluster i / selected from it
$M = \sum_{i=1}^{N} M_i$, $\bar{M} = M/N$	population size and average cluster size
y_{ij}	value of the jth sampled element of cluster i
$\bar{y}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}$	sample mean of cluster i
μ_i, $\tau_i = M_i\mu_i$	true mean and total of cluster i; $\tau = \sum_{i=1}^{N}\tau_i$
$\hat{\tau}_i = M_i\bar{y}_i$	estimated total of cluster i
$S_i^2 = \frac{1}{M_i-1}\sum_{j=1}^{M_i}(y_{ij}-\mu_i)^2$	within-cluster variance
$S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i-\bar{\tau})^2$, $\bar{\tau} = \tau/N$	between-cluster variance of totals

2 Estimation of the Total and the Mean

Theorem 9.2: Unbiased estimators

$$\hat{\tau} = \frac{N}{n}\sum_{i=1}^{n} M_i \bar{y}_i = \frac{N}{n}\sum_{i=1}^{n} M_i \frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}, \qquad \bar{y} = \frac{\hat{\tau}}{M}$$

are unbiased for τ and μ respectively.

Proof. Let E_2 denote expectation over the second stage for a fixed set of clusters, and E_1 over the first stage. For a fixed selected cluster, $\mathrm{E}_2(\bar{y}_i) = \mu_i$ (SRS), so $\mathrm{E}_2(\hat{\tau}_i) = M_i\mu_i = \tau_i$ and

$$\mathrm{E}_2(\hat{\tau}) = \frac{N}{n}\sum_{i=1}^{n} \tau_i.$$

This is N times an SRS mean of the cluster totals, so $\mathrm{E}_1\mathrm{E}_2(\hat{\tau}) = N \cdot \frac{\tau}{N} = \tau$. Then $\mathrm{E}(\bar{y}) = \tau/M = \mu$. ■

Theorem 9.3: Variance

With $u_i = M_i^2 \frac{M_i - m_i}{M_i} \frac{S_i^2}{m_i}$ (the second-stage variance of $\hat{\tau}_i$),

$$\mathrm{Var}(\hat{\tau}) = N^2 \frac{N-n}{N}\frac{S_b^2}{n} + \frac{N}{n}\sum_{i=1}^{N} u_i = \underbrace{N^2 \frac{N-n}{N}\frac{S_b^2}{n}}_{\text{between psu}} + \underbrace{\frac{N}{n}\sum_{i=1}^{N} M_i^2 \frac{M_i - m_i}{M_i}\frac{S_i^2}{m_i}}_{\text{within psu}}.$$

2.1 Variance of a two-stage estimator

Theorem 9.4: Conditional variance decomposition

For any estimator $\hat{\theta}$ computed from a two-stage sample,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}_1\big(\mathrm{E}_2(\hat{\theta})\big) + \mathrm{E}_1\big(\mathrm{Var}_2(\hat{\theta})\big).$$

For three stages, $\mathrm{Var}(\hat{\theta}) = \mathrm{Var}_1(\mathrm{E}_2\mathrm{E}_3\hat{\theta}) + \mathrm{E}_1\big(\mathrm{Var}_2(\mathrm{E}_3\hat{\theta})\big) + \mathrm{E}_1\mathrm{E}_2\big(\mathrm{Var}_3\hat{\theta}\big)$.

Proof. Let $\theta = \mathrm{E}(\hat{\theta})$. Conditionally on the first stage,

$$\mathrm{E}_2(\hat{\theta} - \theta)^2 = \mathrm{Var}_2(\hat{\theta}) + \big(\mathrm{E}_2\hat{\theta}\big)^2 - 2\theta\,\mathrm{E}_2\hat{\theta} + \theta^2.$$

Taking E_1 and using $\mathrm{E}_1\mathrm{E}_2\hat{\theta} = \theta$,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{E}_1\mathrm{Var}_2(\hat{\theta}) + \mathrm{E}_1\big(\mathrm{E}_2\hat{\theta}\big)^2 - \theta^2 = \mathrm{E}_1\mathrm{Var}_2(\hat{\theta}) + \mathrm{Var}_1\big(\mathrm{E}_2\hat{\theta}\big).$$

■

Derivation: proof of the variance formula

Between term. $\mathrm{E}_2(\hat{\tau}) = \frac{N}{n}\sum_{i=1}^{n}\tau_i$ is N times the SRS mean of the N cluster totals, so by Ch. 2

$$\mathrm{Var}_1\big(\mathrm{E}_2\hat{\tau}\big) = N^2\frac{N-n}{N}\frac{S_b^2}{n}.$$

Within term. Given the clusters, the subsamples are independent, and $\mathrm{Var}_2(\hat\tau_i) = M_i^2 \mathrm{Var}_2(\bar y_i) = M_i^2 \frac{M_i - m_i}{M_i}\frac{S_i^2}{m_i} = u_i$. Thus

$$\mathrm{Var}_2(\hat\tau) = \frac{N^2}{n^2}\sum_{i=1}^{n} u_i, \qquad \mathrm{E}_1\mathrm{Var}_2(\hat\tau) = \frac{N^2}{n^2}\cdot n \cdot \frac{1}{N}\sum_{i=1}^{N} u_i = \frac{N}{n}\sum_{i=1}^{N} u_i,$$

because each cluster is in the first-stage sample with probability n/N.
Direct expansion (alternative). Writing $\hat\tau - \tau = \big(\frac{N}{n}\sum_{i\le n}\tau_i - \tau\big) + \frac{N}{n}\sum_{i\le n}(\hat\tau_i - \tau_i)$ and squaring, the cross term has $\mathrm{E}_2(\hat\tau_i - \tau_i) = 0$, and the products $(\hat\tau_i - \tau_i)(\hat\tau_{i'} - \tau_{i'})$, $i \neq i'$, have zero conditional expectation since $\mathrm{E}_2(\hat\tau_i\hat\tau_{i'}) = \tau_i\tau_{i'}$ by independence of the subsamples. The two remaining squared terms give the between and within components.

2.2 Estimated variance

Theorem 9.5: Unbiased variance estimator

$$\widehat{\mathrm{Var}}(\hat\tau) = N^2\frac{N-n}{N}\frac{s_b^2}{n} + \frac{N}{n}\sum_{i=1}^{n} M_i^2\frac{M_i - m_i}{M_i}\frac{s_i^2}{m_i},$$

$$s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\bar y_i - \bar M\bar y)^2, \qquad s_i^2 = \frac{1}{m_i - 1}\sum_{j=1}^{m_i}(y_{ij} - \bar y_i)^2,$$

and $\widehat{\mathrm{Var}}(\bar y) = \widehat{\mathrm{Var}}(\hat\tau)/M^2$. Here s_b^2 is the sample variance of the $\hat\tau_i = M_i\bar y_i$ (note $\bar M\bar y = \hat\tau/N = \frac{1}{n}\sum\hat\tau_i$).

The key fact is that s_b^2 is *not* unbiased for S_b^2: the estimated cluster totals also carry second-stage noise.

Theorem 9.6

$\mathrm{E}(s_b^2) = S_b^2 + \frac{1}{N}\sum_{i=1}^{N} u_i$, and consequently $\mathrm{E}[\widehat{\mathrm{Var}}(\hat\tau)] = \mathrm{Var}(\hat\tau)$.

Derivation: proof

Write $\bar{\hat\tau} = \frac{1}{n}\sum_{i\le n}\hat\tau_i = \hat\tau/N$. Then $(n-1)s_b^2 = \sum_{i\le n}\hat\tau_i^2 - n\bar{\hat\tau}^2$.
First piece. $\mathrm{E}_2(\hat\tau_i^2) = \mathrm{Var}_2(\hat\tau_i) + \tau_i^2 = u_i + \tau_i^2$, so $\mathrm{E}\big[\sum_{i\le n}\hat\tau_i^2\big] = \frac{n}{N}\sum_{i=1}^{N}(u_i + \tau_i^2)$.
Second piece. $\mathrm{E}(\bar{\hat\tau}^2) = \frac{\mathrm{Var}(\hat\tau)}{N^2} + \bar\tau^2 = \frac{N-n}{Nn}S_b^2 + \frac{1}{nN}\sum_{i=1}^{N} u_i + \bar\tau^2$.
Combining and using $\frac{1}{N}\sum_{i=1}^{N}\tau_i^2 - \bar\tau^2 = \frac{N-1}{N}S_b^2$,

$$\begin{aligned}\mathrm{E}\big[(n-1)s_b^2\big] &= \frac{n}{N}\sum u_i + n\Big(\frac{1}{N}\sum\tau_i^2 - \bar\tau^2\Big) - \frac{N-n}{N}S_b^2 - \frac{1}{N}\sum u_i \\ &= \frac{n-1}{N}\sum_{i=1}^{N} u_i + \frac{n(N-1) - (N-n)}{N}S_b^2 = (n-1)\Big[S_b^2 + \frac{1}{N}\sum_{i=1}^{N} u_i\Big].\end{aligned}$$

Unbiasedness of $\widehat{\mathrm{Var}}(\hat\tau)$. Since $\mathrm{E}_2(s_i^2) = S_i^2$, the second term has expectation $\frac{N}{n}\cdot\frac{n}{N}\sum_{i=1}^{N} u_i = \sum_{i=1}^{N} u_i$. Hence

$$\mathrm{E}\widehat{\mathrm{Var}}(\hat\tau) = N^2\frac{N-n}{Nn}\Big(S_b^2 + \frac{1}{N}\sum u_i\Big) + \sum u_i = N^2\frac{N-n}{Nn}S_b^2 + \Big(\frac{N-n}{n} + 1\Big)\sum u_i = \mathrm{Var}(\hat\tau).$$

Of the within-psu variance $\frac{N}{n}\sum u_i$, the between-psu term of $\widehat{\text{Var}}$ already contributes $\frac{N-n}{n}\sum u_i$, and the explicit within term supplies the remaining $\sum u_i$.

Remark 9.7: Special cases

- All elements of each selected psu are measured ($m_i = M_i$): $\widehat{\text{Var}}(\hat{\tau}) = N^2\frac{N-n}{Nn}s_b^2$, which is one-stage cluster sampling (Ch. 8).
- All psu are included ($n = N$): $\widehat{\text{Var}}(\hat{\tau}) = \sum_{i=1}^{N} M_i^2 \frac{M_i - m_i}{M_i}\frac{s_i^2}{m_i}$, which is stratified random sampling with the clusters as strata (Ch. 3).

3 Estimating the Mean when M is Unknown

$\bar{y} = \hat{\tau}/M$ requires M. Otherwise use the ratio estimator

$$\hat{\mu} = \frac{\sum_{i=1}^{n} M_i\bar{y}_i}{\sum_{i=1}^{n} M_i} = \frac{\sum_{i=1}^{n}\hat{\tau}_i}{\sum_{i=1}^{n} M_i}.$$

Theorem 9.8

$$\widehat{\text{Var}}(\hat{\mu}) = \frac{N^2}{M^2}\cdot\frac{N-n}{N}\cdot\frac{s_r^2}{n} + \frac{N}{nM^2}\sum_{i=1}^{n} M_i^2\frac{M_i - m_i}{M_i}\frac{s_i^2}{m_i}, \qquad s_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\bar{y}_i - M_i\hat{\mu})^2,$$

where, when M is unknown, M is estimated by $\hat{M} = N\cdot\frac{1}{n}\sum_{i=1}^{n} M_i$ (equivalently $\bar{M}$ by the sample average of the M_i).

The only change from $\widehat{\text{Var}}(\bar{y})$ is that the between term uses deviations $M_i\bar{y}_i - M_i\hat{\mu}$ (residuals from a ratio fit) instead of $M_i\bar{y}_i - \bar{M}\bar{y}$.

4 Estimation of a Proportion

For the proportion of students in favour of a law, let $y_{ij} = 1$ or 0 and $\hat{p}_i = \frac{1}{m_i}\sum_j y_{ij}$, the sample proportion in cluster i. With M unknown,

$$\hat{p} = \frac{\sum_{i=1}^{n} M_i\hat{p}_i}{\sum_{i=1}^{n} M_i},$$

$$\widehat{\text{Var}}(\hat{p}) = \frac{N^2}{M^2}\frac{N-n}{N}\frac{s_p^2}{n} + \frac{N}{nM^2}\sum_{i=1}^{n} M_i^2\frac{M_i - m_i}{M_i}\frac{\hat{p}_i\hat{q}_i}{m_i - 1}, \qquad s_p^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\hat{p}_i - M_i\hat{p})^2,$$

with $\hat{q}_i = 1 - \hat{p}_i$. The within term uses the SRS result $\widehat{\text{Var}}(\hat{p}_i) = \frac{\hat{p}_i\hat{q}_i}{m_i - 1}\frac{M_i - m_i}{M_i}$.

5 Equal-Sized Clusters

Suppose $M_1 = \cdots = M_N = \bar{M}$ and $m_1 = \cdots = m_n = m$. Then

$$\bar{y} = \frac{\hat{\tau}}{M} = \frac{1}{N\bar{M}}\cdot\frac{N}{n}\bar{M}\sum_{i=1}^{n}\bar{y}_i = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i = \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} y_{ij} = \hat{\mu},$$

the overall sample average. Arrange the data as an $m \times n$ table (columns = psu) with

$$MSB = \frac{m}{n-1}\sum_{i=1}^{n}(\bar{y}_i - \bar{y})^2, \qquad MSW = \frac{1}{n(m-1)}\sum_{i=1}^{n}\sum_{j=1}^{m}(y_{ij} - \bar{y}_i)^2.$$

Theorem 9.9: Equal-size two-stage variance

With $f_1 = n/N$ and $f_2 = m/\bar{M}$,

$$\widehat{\mathrm{Var}}(\bar{y}) = (1-f_1)\frac{MSB}{nm} + (1-f_2)\frac{MSW}{Nm}.$$

If N is large, $\widehat{\mathrm{Var}}(\bar{y}) \approx MSB/(nm)$.

Derivation

Between term. $s_b^2 = \frac{1}{n-1}\sum(\bar{M}\bar{y}_i - \bar{M}\bar{y})^2 = \bar{M}^2\frac{MSB}{m}$, and $N^2/M^2 = 1/\bar{M}^2$, so

$$\frac{N^2}{M^2}\frac{N-n}{N}\frac{s_b^2}{n} = (1-f_1)\frac{1}{n}\frac{MSB}{m}.$$

Within term. $\frac{1}{n}\sum_i s_i^2 = MSW$, so

$$\frac{1}{M^2}\frac{N}{n}\sum_{i=1}^{n}\bar{M}^2\frac{\bar{M}-m}{\bar{M}}\frac{s_i^2}{m} = \frac{N\bar{M}^2}{M^2}(1-f_2)\frac{MSW}{m} = \frac{1}{N}(1-f_2)\frac{MSW}{m}.$$

The same computation for $\widehat{\mathrm{Var}}(\hat{\mu})$ gives $s_r^2 = s_b^2$ and $\hat{M} = M$, so $\widehat{\mathrm{Var}}(\hat{\mu}) = \widehat{\mathrm{Var}}(\bar{y})$ here.

Remark 9.10: Limiting cases

- $m = \bar{M}$ ($f_2 = 1$): the within term vanishes and $\widehat{\mathrm{Var}}(\bar{y}) = (1-f_1)\frac{MSB}{nm}$, one-stage cluster sampling.
- $n = N$ ($f_1 = 1$): $\widehat{\mathrm{Var}}(\bar{y}) = (1-f_2)\frac{MSW}{Nm}$, the variance estimate of a stratified random sample with N strata and m observations per stratum.

6 Probabilities Proportional to Size

PPS sampling is useful when cluster sizes vary greatly; it is typically used only at the first stage. Selecting psu with replacement with probabilities $z_i = M_i/M$ and estimating each cluster total by $\hat{\tau}_i = M_i\bar{y}_i$,

$$\hat{\tau}_{\mathrm{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{\hat{\tau}_i}{z_i} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\mathrm{pps}} = \frac{\hat{\tau}_{\mathrm{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i,$$

compared with the equal-probability choice $z_i = 1/N$, which gives $\hat{\tau} = \frac{N}{n}\sum M_i\bar{y}_i$. The variance estimators are

$$\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2,$$

and they already include the second-stage variability, since each $\bar{y}_i$ is an independent draw of a cluster *and* a subsample. With $m_i = m$ we get $\hat{\mu}_{\mathrm{pps}} = \bar{y}$ and $\widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}}) = \frac{1}{n(n-1)}\sum(\bar{y}_i - \bar{y})^2 = \frac{MSB}{nm}$, the large-$N$ approximation of the previous section.

Chapter 10. Estimating Population Size and Nonresponse

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1 Capture–Recapture Estimation of N

To estimate the number N of fish in a lake, catch n_1 fish, mark them and release them; later catch a second sample of n_2 fish and count the number m of marked fish among them.

Definition 10.1: Assumptions

1. The population is closed (no births, deaths or migration between the samples).
2. Each sample is an SRS from the population (marking does not change catchability).
3. The two samples are independent.

The proportion of marked fish in the second sample should resemble that in the population:

$$\frac{m}{n_2} \approx \frac{n_1}{N} \quad \Longrightarrow \quad \hat{N} = \frac{n_1 n_2}{m}$$

(the Lincoln–Petersen estimator), where n_1 is the size of the first (capture) sample, n_2 that of the second (recapture) sample, and m the number of marked elements in the second sample.

1.1 Capture–recapture as a ratio estimator

Theorem 10.2

$\hat{N}$ is a ratio estimator of a population total, and

$$\widehat{\mathrm{Var}}(\hat{N}) = \frac{n_1^2 n_2^2 (n_2 - m)}{m^3 (n_2 - 1)} \approx \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

Define for every element $y_i = 1$, so $\tau_y = \sum_{i=1}^{N} y_i = N$, and $x_i = 1$ if element i is marked, 0 otherwise, so $\tau_x = \sum_{i=1}^{N} x_i = n_1$ is known. The recapture sample of size n_2 gives

$$r = \frac{\sum_{i=1}^{n_2} y_i}{\sum_{i=1}^{n_2} x_i} = \frac{n_2}{m}, \qquad \hat{N} = \hat{\tau}_y = r\tau_x = \frac{n_2}{m} n_1.$$

From Ch. 5, $\widehat{\mathrm{Var}}(\hat{N}) = \tau_x^2 \widehat{\mathrm{Var}}(r) = n_1^2 \widehat{\mathrm{Var}}(r)$ with

$$\widehat{\mathrm{Var}}(r) = \frac{1}{\bar{x}^2} \cdot \frac{N - n_2}{N} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (y_i - r x_i)^2,$$

and we take the fpc ≈ 1. Now

$$\bar{x} = \frac{m}{n_2}, \qquad \sum_{i=1}^{n_2} (y_i - r x_i)^2 = \sum y_i^2 - 2r \sum x_i y_i + r^2 \sum x_i^2 = n_2 - 2\frac{n_2}{m} m + \frac{n_2^2}{m^2} m = \frac{n_2(n_2 - m)}{m},$$

using $y_i^2 = y_i = 1$ and $x_i^2 = x_i$. Therefore

$$\widehat{\mathrm{Var}}(r) = \frac{n_2^2}{m^2} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \cdot \frac{n_2(n_2 - m)}{m} = \frac{n_2^2 (n_2 - m)}{m^3 (n_2 - 1)},$$

and multiplying by n_1^2 gives the result.

1.2 Maximum likelihood

Definition 10.3

Let $f(\mathbf{x} \mid \theta)$ be the joint pdf of the sample $\mathbf{X} = (X_1, \ldots, X_n)$. Given $\mathbf{X} = \mathbf{x}$, the function $L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)$ of θ is the **likelihood function**. If $L(\theta_1 \mid \mathbf{x}) \geq L(\theta_2 \mid \mathbf{x})$, then θ_1 is more plausible than θ_2 in light of the data. The **maximum likelihood estimator** is $\hat{\theta}(\mathbf{x}) = \arg\max_{\theta \in \Theta} L(\theta \mid \mathbf{x})$.

Example 10.4: Normal mean

For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$,

$$\ell(\theta) = \log L(\theta \mid \mathbf{x}) = -\frac{n}{2} \log 2\pi - \frac{1}{2} \sum_{i=1}^{n} (x_i - \theta)^2, \qquad \ell'(\hat{\theta}) = \sum (x_i - \hat{\theta}) = 0 \;\Rightarrow\; \hat{\theta} = \bar{x}.$$

Theorem 10.5: Large-sample properties of the MLE

Under regularity conditions,

1. (consistency) $\lim_{n \to \infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0$ for every $\varepsilon > 0$;
2. (asymptotic normality) $\sqrt{n}(\hat{\theta}_n - \theta) \to N\big(0, 1/I(\theta)\big)$, i.e. $\hat{\theta}_n \approx N\big(\theta, \frac{1}{nI(\theta)}\big)$, where $I(\theta) = \mathrm{E}\big[\big(\frac{\partial}{\partial \theta} \log f(X \mid \theta)\big)^2\big]$ is the Fisher information of one observation.

Theorem 10.6: MLE of N

Model the recapture sample as $X_i = 1$ (marked) or 0, $X_i \overset{\text{iid}}{\sim} \mathrm{Ber}(p)$ with $p = n_1/N$, $i = 1, \ldots, n_2$. Then

$$\hat{N}_{\mathrm{MLE}} = \frac{n_1 n_2}{m}, \qquad \mathrm{Var}(\hat{N}_{\mathrm{MLE}}) \approx \frac{N^2 (N - n_1)}{n_1 n_2}, \qquad \widehat{\mathrm{Var}}(\hat{N}_{\mathrm{MLE}}) = \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

With $\sum_{i=1}^{n_2} X_i = m$,

$$L(N) = \prod_{i=1}^{n_2} \Big(\frac{n_1}{N}\Big)^{X_i}\Big(1-\frac{n_1}{N}\Big)^{1-X_i} = \Big(\frac{n_1}{N}\Big)^{m}\Big(1-\frac{n_1}{N}\Big)^{n_2-m},$$

$$\ell(N) = m\log n_1 - m\log N + (n_2-m)\log(N-n_1) - (n_2-m)\log N, \qquad \ell'(N) = \frac{n_2-m}{N-n_1} - \frac{n_2}{N}.$$

Setting $\ell'(N)=0$: $N(n_2-m) = n_2(N-n_1)$, so $\hat N = n_1 n_2/m$.

Fisher information. For one observation,

$$\frac{\partial}{\partial N}\log f(x\mid N) = \frac{1-x}{N-n_1} - \frac{1}{N} = \frac{n_1/N - x}{N-n_1} = \frac{p-x}{N-n_1},$$

$$I(N) = \mathrm{E}\Big(\frac{p-X}{N-n_1}\Big)^2 = \frac{p(1-p)}{(N-n_1)^2} = \frac{n_1}{N^2(N-n_1)}.$$

Hence $\mathrm{Var}(\hat N_{\mathrm{MLE}}) \approx \big[n_2 I(N)\big]^{-1} = \frac{N^2(N-n_1)}{n_1 n_2}$. Plugging in $\hat N = n_1 n_2/m$:

$$\frac{(n_1n_2/m)^2\,(n_1n_2/m - n_1)}{n_1n_2} = \frac{n_1^2n_2^2}{m^2}\cdot\frac{n_1(n_2-m)}{m}\cdot\frac{1}{n_1n_2} = \frac{n_1^2 n_2(n_2-m)}{m^3}.$$

The ratio-estimator and likelihood approaches agree: $\widehat{\mathrm{Var}}(\hat\tau_y) \approx \widehat{\mathrm{Var}}(\hat N_{\mathrm{MLE}})$.

Example 10.7: Illustration

If $n_1 = 200$ fish are marked and a second catch of $n_2 = 100$ contains $m = 20$ marked fish, then $\hat N = 200\cdot 100/20 = 1000$ and

$$\widehat{\mathrm{Var}}(\hat N) = \frac{200^2\cdot 100\cdot 80}{20^3} = 40{,}000, \qquad 2\sqrt{40000} = 400,$$

so N is estimated as 1000 ± 400.

2 Nonresponse

Nonresponse (missing data) biases the estimation of population parameters. Divide the population into a response stratum R and a nonresponse stratum:

	size	mean
Respondents	N_1	μ_1
Nonrespondents	N_2	μ_2
	$N = N_1 + N_2$	μ

Theorem 10.8: Nonresponse bias

If $\bar y_1$ is unbiased for the respondents' mean μ_1, then as an estimator of μ,

$$\mathrm{E}(\bar y_1) - \mu = \frac{N_2}{N}(\mu_1 - \mu_2).$$

Proof. $\mu = \frac{N_1}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \frac{N_1+N_2}{N}\mu_1 - \frac{N_2}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \mu_1 - \frac{N_2}{N}(\mu_1-\mu_2)$. ∎

The bias is small only if the nonresponse rate N_2/N is small or respondents and nonrespondents are alike ($\mu_1 \approx \mu_2$). Increasing the sample size does not reduce it.

2.1 A conservative interval for a proportion

Suppose n_1 respondents are an SRS from the N_1 respondents of the population, so a 95% interval for their proportion p_1 is

$$\hat{p}_1 \pm 2\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1 - 1} \cdot \frac{N_1 - n_1}{N_1}}.$$

Nothing is known about the nonrespondents' proportion p_2, except $0 \le p_2 \le 1$. With $w_1 = N_1/N$ and $w_2 = N_2/N$, $p = w_1 p_1 + w_2 p_2$. Take the worst cases $p_2 = 0$ for the lower limit and $p_2 = 1$ for the upper limit:

$$\hat{p}_L = w_1\Big(\hat{p}_1 - 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\Big) + w_2 \cdot 0,$$
$$\hat{p}_U = w_1\Big(\hat{p}_1 + 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\Big) + w_2 \cdot 1.$$

Whenever the interval for p_1 covers p_1, $[\hat{p}_L, \hat{p}_U]$ covers p whatever p_2 is, so $P(\hat{p}_L \le p \le \hat{p}_U) \ge 0.95$. The width grows by w_2: a high nonresponse rate makes the interval uninformative, which is why reducing nonresponse (call-backs, follow-ups) matters more than increasing n.

Chapter 11. Resampling Methods

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1 Training Error and Test Error

Resampling methods repeatedly draw samples from the data at hand to learn about an estimator. The two most common are **cross-validation** and the **bootstrap**; they estimate the prediction error, the standard deviation and the bias of estimates.

Definition 11.1

Let $D_n = \{(x_i, y_i)\}_{i=1}^n$ be the training data, g_{D_n} a prediction rule fitted to it, and L a loss function.

- The **training error** applies the fitted rule back to the training observations, e.g. $\frac{1}{n}\sum(y_i - \hat{y}_i)^2$.
- The **test error** (prediction error) is the expected loss on a new observation (x_0, y_0) independent of D_n:

$$PE = \mathrm{E}\big[L(y_0, g_{D_n}(x_0))\big].$$

The training error is easy to compute but usually underestimates the test error, because the rule was tuned to the same data: a curve that zig-zags through every training point has zero training error yet predicts new points $y_i^* = f(x_i) + \varepsilon_i^*$ badly. The smaller the test error, the more suitable the model. Measures such as $MS_E = \frac{SS_E}{n-2}$ describe goodness of fit, not predictive ability.

With an independent test set $\{(x_j, y_j)\}_{j=1}^{n_t}$ we could estimate

$$\widehat{PE} = \frac{1}{n_t}\sum_{j=1}^{n_t} L(y_j, \hat{y}_j), \qquad \hat{y}_j = g_{D_n}(x_j),$$

but a large test set is often unavailable. The methods below hold out part of the training data instead.

2 Cross-Validation

2.1 K-fold cross-validation

Randomly divide the data into K roughly equal parts $C_1, \dots, C_K$ (C_k holds the indices of part k, n_k observations; $n_k = n/K$ if n is a multiple of K). For each k, fit the model to the other $K-1$ parts and predict part k:

$$\mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in C_k} (y_i - \hat{y}_i)^2, \qquad CV_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k,$$

where $\hat{y}_i$ is the fit for observation i obtained with part k removed.

2.2 Leave-one-out cross-validation

$K = n$ gives LOOCV. Let $D_{(i)}$ be the training set with (x_i, y_i) omitted and $\hat{y}_{(i)} = g_{D_{(i)}}(x_i)$. Then

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} L(y_i, \hat{y}_{(i)}),$$

which mimics $\widehat{PE}$ by letting each point serve as a test point for a rule that did not see it.

Advantages of LOOCV over a single validation split: (i) much less bias, since each fit uses $n-1$ observations, almost the whole data set; (ii) no randomness, since repeated runs give the same result.

Theorem 11.2: LOOCV shortcut for least squares

For least-squares linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$,

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2,$$

where $\hat{y}_i$ is the fitted value from the full fit and h_{ii} is the leverage of observation i. The terms $(y_i - \hat{y}_i)/(1 - h_{ii}) = y_i - \hat{y}_{(i)}$ are the **PRESS residuals**; only one fit is needed.

Derivation: proof

Let $\mathbf{x}_i^\top$ be the ith row of $\mathbf{X}$, $A = \mathbf{X}^\top \mathbf{X}$, and $h_{ii} = \mathbf{x}_i^\top A^{-1} \mathbf{x}_i$. Deleting row i gives $A_{(i)} = A - \mathbf{x}_i \mathbf{x}_i^\top$ and, by the Sherman–Morrison formula,

$$A_{(i)}^{-1} = A^{-1} + \frac{A^{-1} \mathbf{x}_i \mathbf{x}_i^\top A^{-1}}{1 - h_{ii}}.$$

With $\mathbf{X}_{(i)}^\top \mathbf{y}_{(i)} = \mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i$,

$$\hat{y}_{(i)} = \mathbf{x}_i^\top A_{(i)}^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) = \left(\mathbf{x}_i^\top A^{-1} + \frac{h_{ii} \mathbf{x}_i^\top A^{-1}}{1 - h_{ii}} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) = \frac{\mathbf{x}_i^\top A^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i)}{1 - h_{ii}}$$
$$= \frac{\hat{y}_i - h_{ii} y_i}{1 - h_{ii}}.$$

Hence $y_i - \hat{y}_{(i)} = \dfrac{(1 - h_{ii}) y_i - \hat{y}_i + h_{ii} y_i}{1 - h_{ii}} = \dfrac{y_i - \hat{y}_i}{1 - h_{ii}}$.

Remark 11.3: Choice of K

LOOCV does not shake up the data enough: the n fits are trained on nearly identical sets, so their errors are highly correlated and their average can have high variance. $K = 5$ or 10 is usually a better choice.

2.3 Why cross-validation works: model selection

Example 11.4: Choosing the degree of a polynomial

Consider $y_i = f(x_i) + \varepsilon_i$ fitted by $f_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p$, and let $\hat{f}_p$ be its LS estimate. The goal is to select p. A natural target is

$$\mathrm{MSE}(p) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}(f(x_i) - \hat{f}_p(x_i))^2,$$

but f is unknown. Consider instead the prediction squared error for new responses $y_i^* = f(x_i) + \varepsilon_i^*$:

$$\begin{aligned} PSE(p) &= \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}(y_i^* - \hat{f}_p(x_i))^2 = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[(f(x_i) - \hat{f}_p(x_i)) + \varepsilon_i^*]^2 \\ &= \mathrm{MSE}(p) + \sigma^2, \end{aligned}$$

because ε_i^* is independent of $\hat{f}_p$ and has mean zero. So minimizing PSE is equivalent to minimizing MSE. Now let

$$CV(p) = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{f}_p^{-i}(x_i))^2,$$

where $\hat{f}_p^{-i}$ is fitted without the ith point. Since y_i is independent of $\hat{f}_p^{-i}$,

$$\mathrm{E}(y_i - \hat{f}_p^{-i}(x_i))^2 = \sigma^2 + \mathrm{E}(f(x_i) - \hat{f}_p^{-i}(x_i))^2,$$

and if $\hat{f}_p^{-i} \approx \hat{f}_p$ (leaving out one point changes little), $\mathrm{E}[CV(p)] \approx \sigma^2 + \mathrm{MSE}(p) = PSE(p)$. Thus $\hat{p} = \arg\min_p CV(p)$ is a sensible rule. The training error $\frac{1}{n}\sum(y_i - \hat{f}_p(x_i))^2$ does not have this property: it keeps decreasing as p grows.

3 The Bootstrap

The bootstrap quantifies the uncertainty of an estimator or a learning method, e.g. the standard error of a coefficient or a confidence interval for it.

3.1 A simple example

We invest a fraction α of our money in an asset with return X and $1-\alpha$ in one with return Y, and choose α to minimize the risk $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

Theorem 11.5

The risk is minimized by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof. $g(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$, and

$$g'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0 \iff \alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}.$$

Since $g'' = 2\mathrm{Var}(X - Y) \geq 0$, this is a minimum. ■

The variances are unknown, so we plug in sample estimates to get $\hat{\alpha}$. To learn the standard deviation of $\hat{\alpha}$ in a simulation, we could draw 1,000 independent data sets of 100 pairs from the population, compute $\hat{\alpha}_1, \ldots, \hat{\alpha}_{1000}$, and report

$$\bar{\alpha} = \frac{1}{1000}\sum_{r=1}^{1000} \hat{\alpha}_r, \qquad \sqrt{\frac{1}{1000-1}\sum_{r=1}^{1000}(\hat{\alpha}_r - \bar{\alpha})^2}.$$

3.2 Back to the real world

For real data we cannot draw new samples from the population. The bootstrap instead draws data sets *from the original data set with replacement*, using the computer to mimic the process of generating new samples.

Definition 11.6: Bootstrap standard error

1. From the original data Z of size n, draw a bootstrap data set Z^{*1} of size n by sampling observations with replacement (some observations appear more than once, some not at all).
2. Compute the estimate $\hat{\alpha}^{*1}$ from Z^{*1}.
3. Repeat B times (say $B = 100$ or 1000) to get $\hat{\alpha}^{*1}, \ldots, \hat{\alpha}^{*B}$.
4. Estimate the standard error by

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \bar{\alpha}^*\right)^2}, \qquad \bar{\alpha}^* = \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}.$$

Example 11.7: Three observations

With $Z = \{(4.3, 2.4), (2.1, 1.1), (5.3, 2.8)\}$ (observations 1, 2, 3), possible bootstrap samples are $Z^{*1} = \{3, 1, 3\}$, $Z^{*2} = \{2, 3, 1\}$, $\ldots$, $Z^{*B} = \{2, 2, 1\}$ (by observation number), each yielding its own $\hat{\alpha}^{*r}$.

In R, `sample(x, replace=T)` draws a bootstrap sample of the indices.